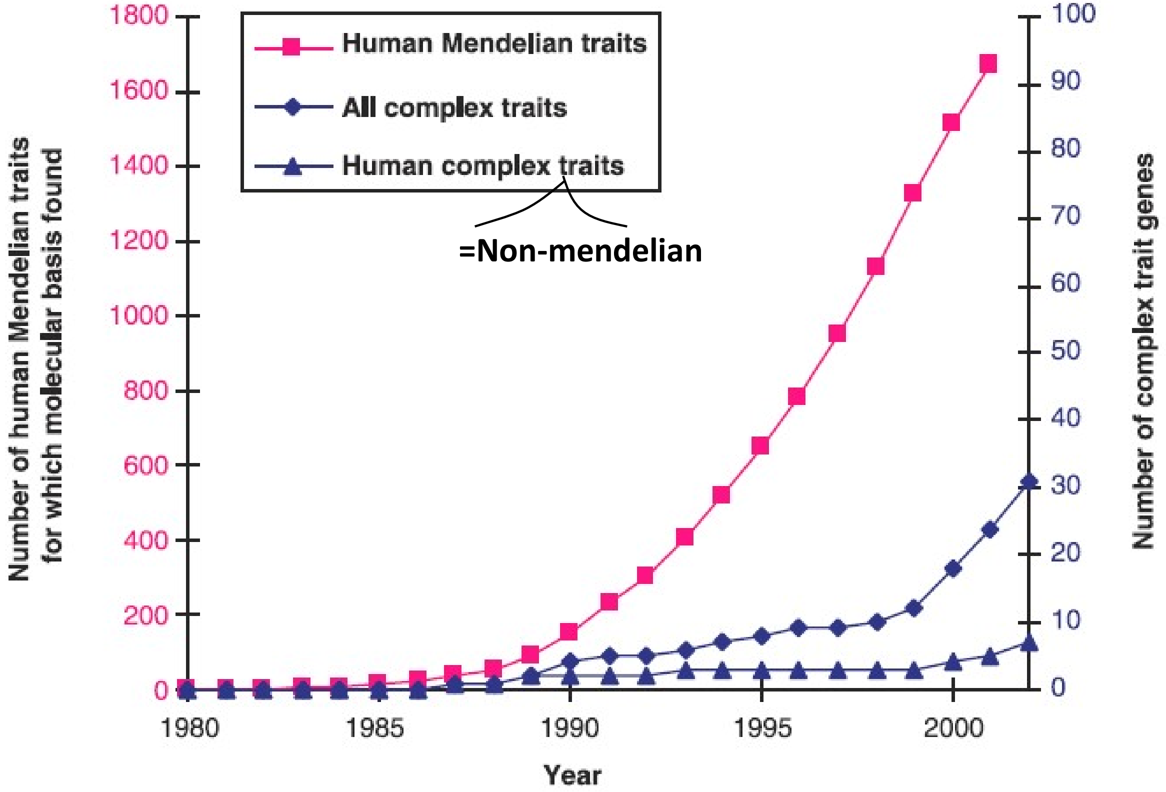
Systems Genetics

Spring 2025

Lecture 5

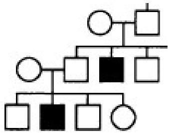
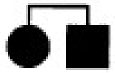
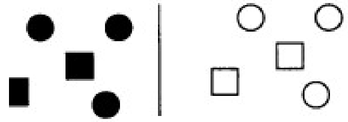
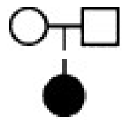
Genetic structure of human population and the HapMap project (GWAS)

Identification of genes underlying human traits



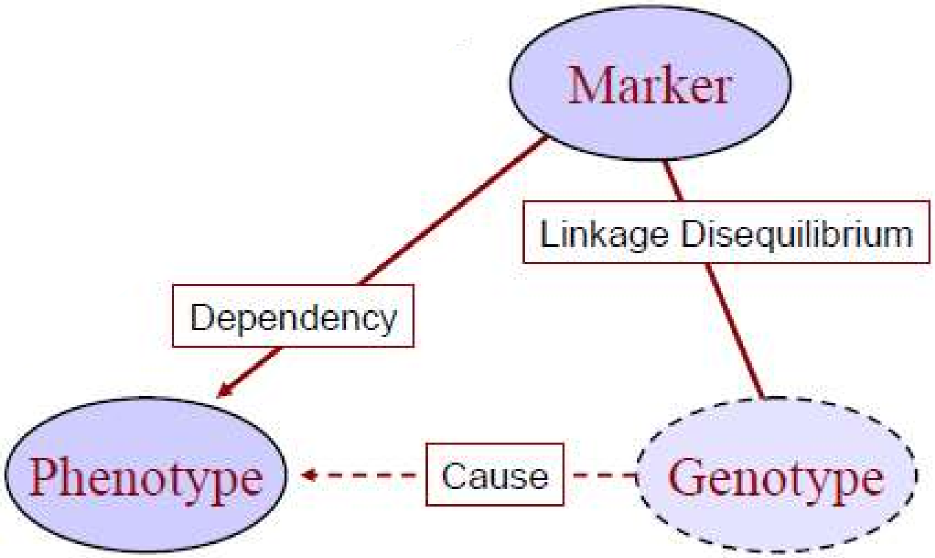
- The candidate gene approach is inadequate: most disease genes are unsuspected on the basis of previous knowledge
- Rare Mendelian disease are almost always caused by rare variants

Glazier et al. science 2002

	linkage	allele sharing	association	TDT
study design	family based 	family based 	population based 	family based 
design complexity	1 high	2 mid-high	4 low	3 mid-low
disease type	Mendelian	Mendelian + complex	Mendelian + complex	Mendelian + complex
method	parametric	a-parametric	a-parametric	a-parametric
population structure problem	no	no	yes	no
Use marker as proxy	yes	yes	? (no for now)	yes (requires association)
quantitative method			regression analysis, ANOVA	
power	best if model can be specified +			

From genotype to phenotype

Task: Find basis (genotype) of diseases (phenotype).
Problem: *Real* genotype is not observable.
But, marker available in linkage with the causal genotype (linkage = the tendency of two markers on the same chromosome to be co-inherited).
Strategy: Use marker as genotype proxy.



Population stratification (cont)
Solution: restricting to a single ancestry.

- Example: association of height in European Americans.

The height trait is associated with the lactase SNP

phenotype

Lactase SNP

Tall

Short

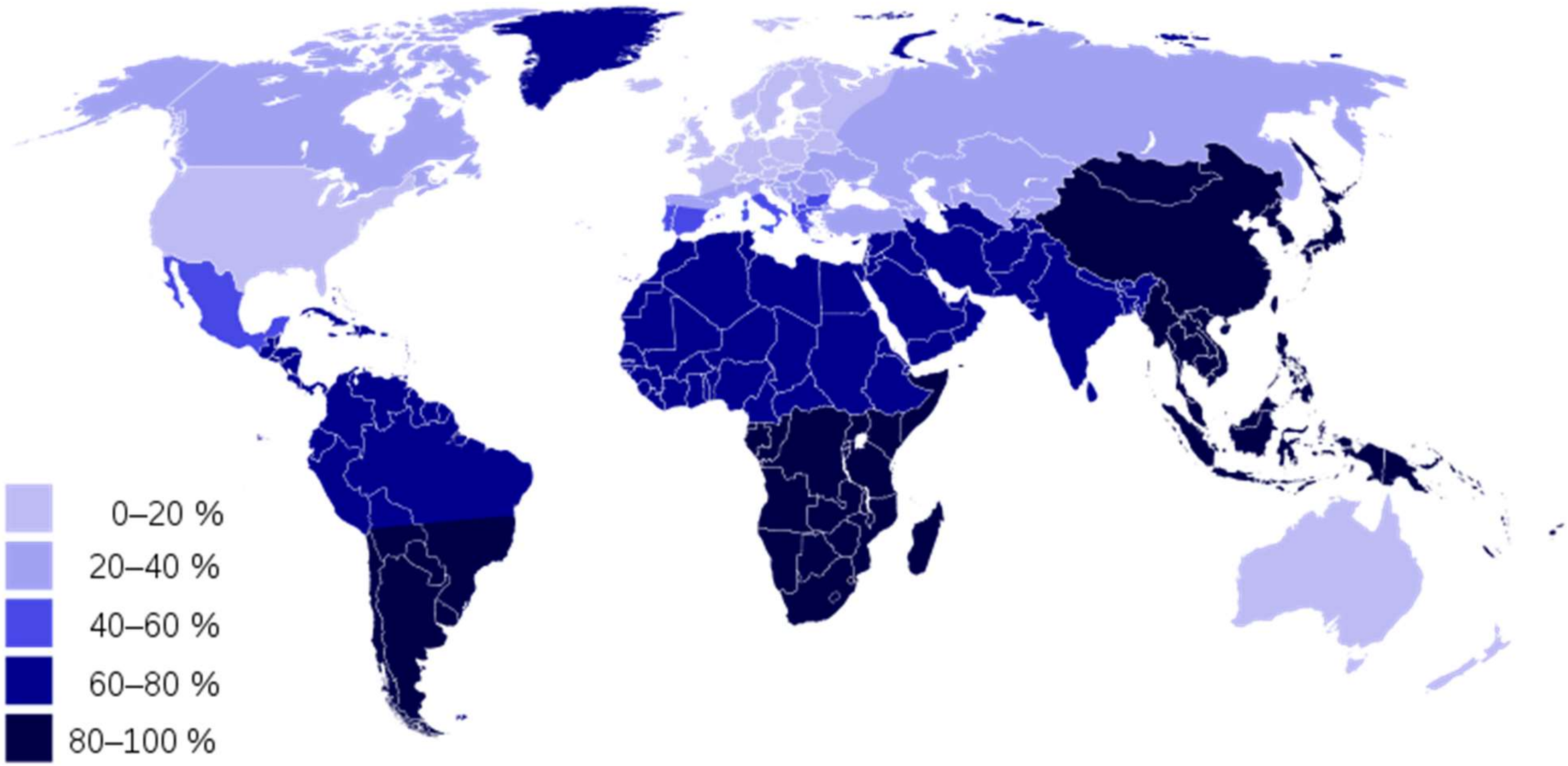
T

C

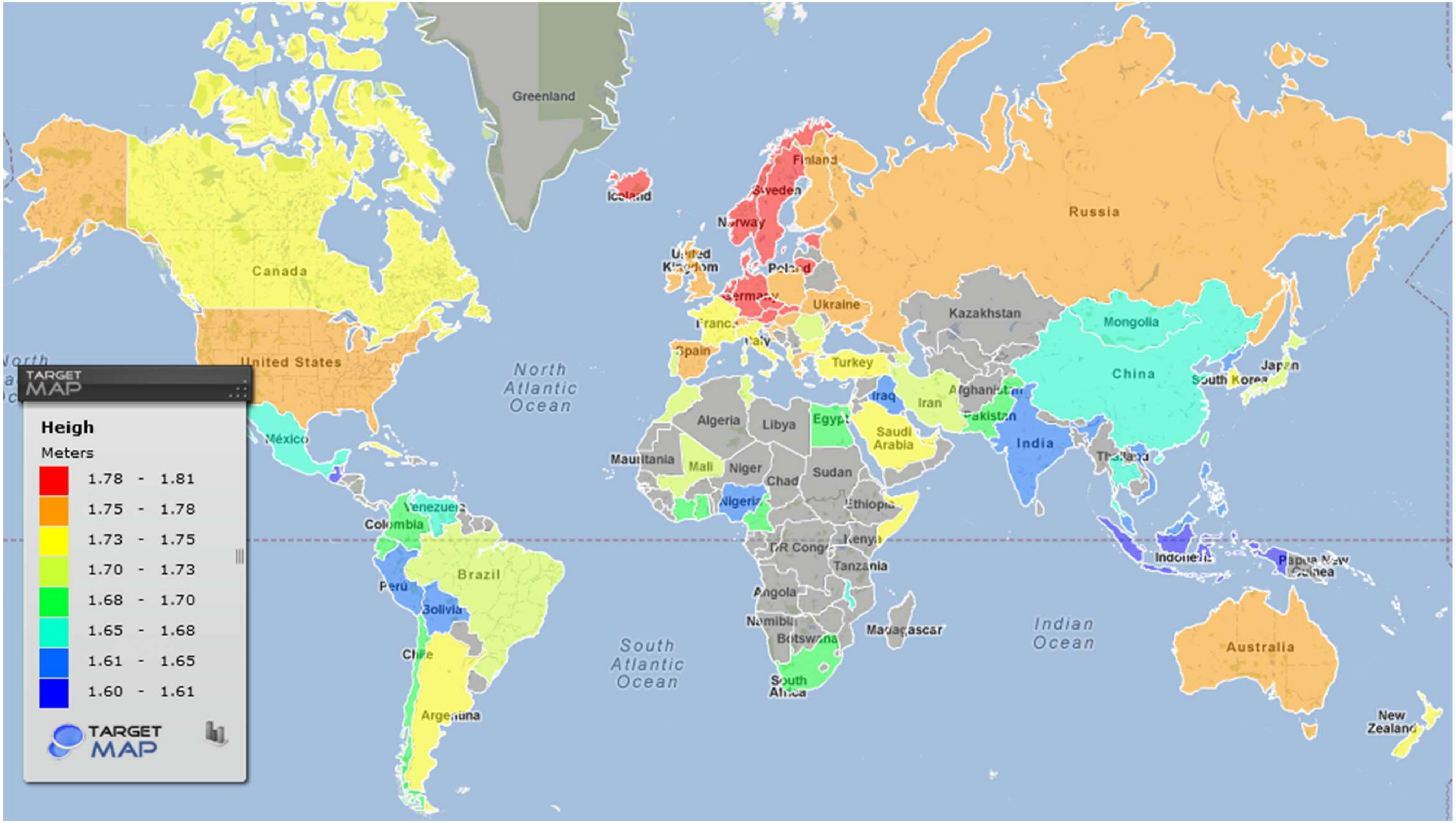
chr2

The same SNP is associated with lactase activity in adults

The frequency of decreased lactase activity around the world



Average Male height map



Population stratification (cont) Solution: restricting to a single ancestry.

- Example: association of height in European Americans.

The height trait is associated with the lactase SNP

phenotype Ancestry Lactase SNP

Tall ⟷ North Europe ⟷ T

Short ⟷ SE Europe ⟷ C

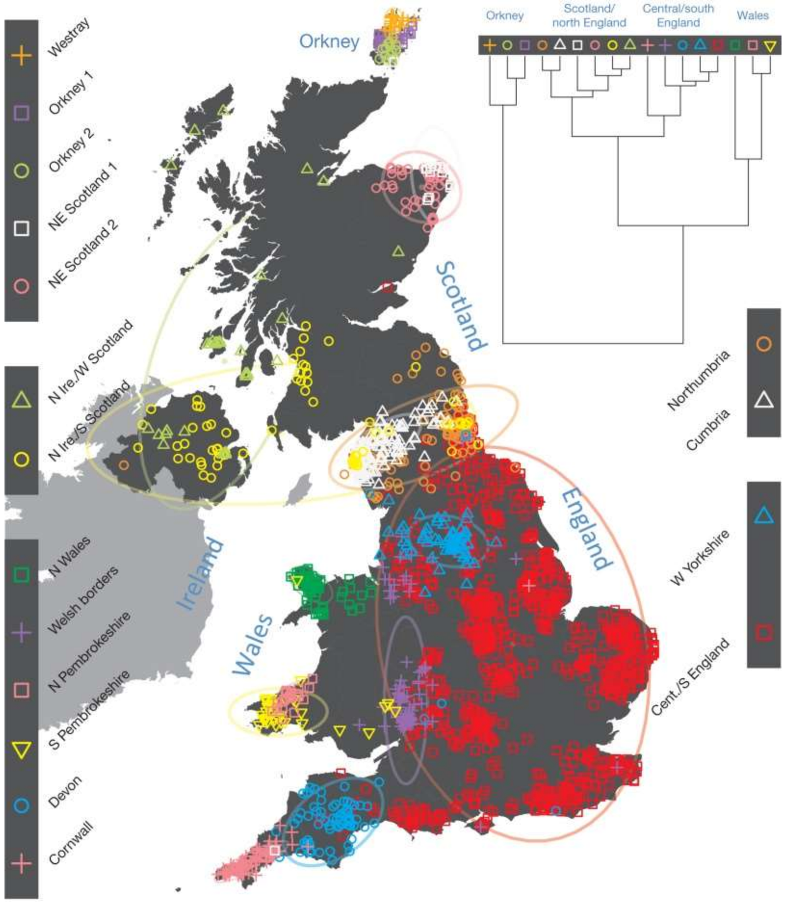
chr2

Spurious association due to stratification!!

@A. price

A rich pattern of subtle fine-scale genetic differentiation within the UK, which shows a marked concordance with geography (Leslie et al. Nature 2015)

Clustering of the 2,039 UK individuals into 17 clusters based only on genetic data.



For each individual, the coloured symbol representing the genetic cluster to which the individual is assigned is plotted at the centroid of their grandparents' birthplaces.

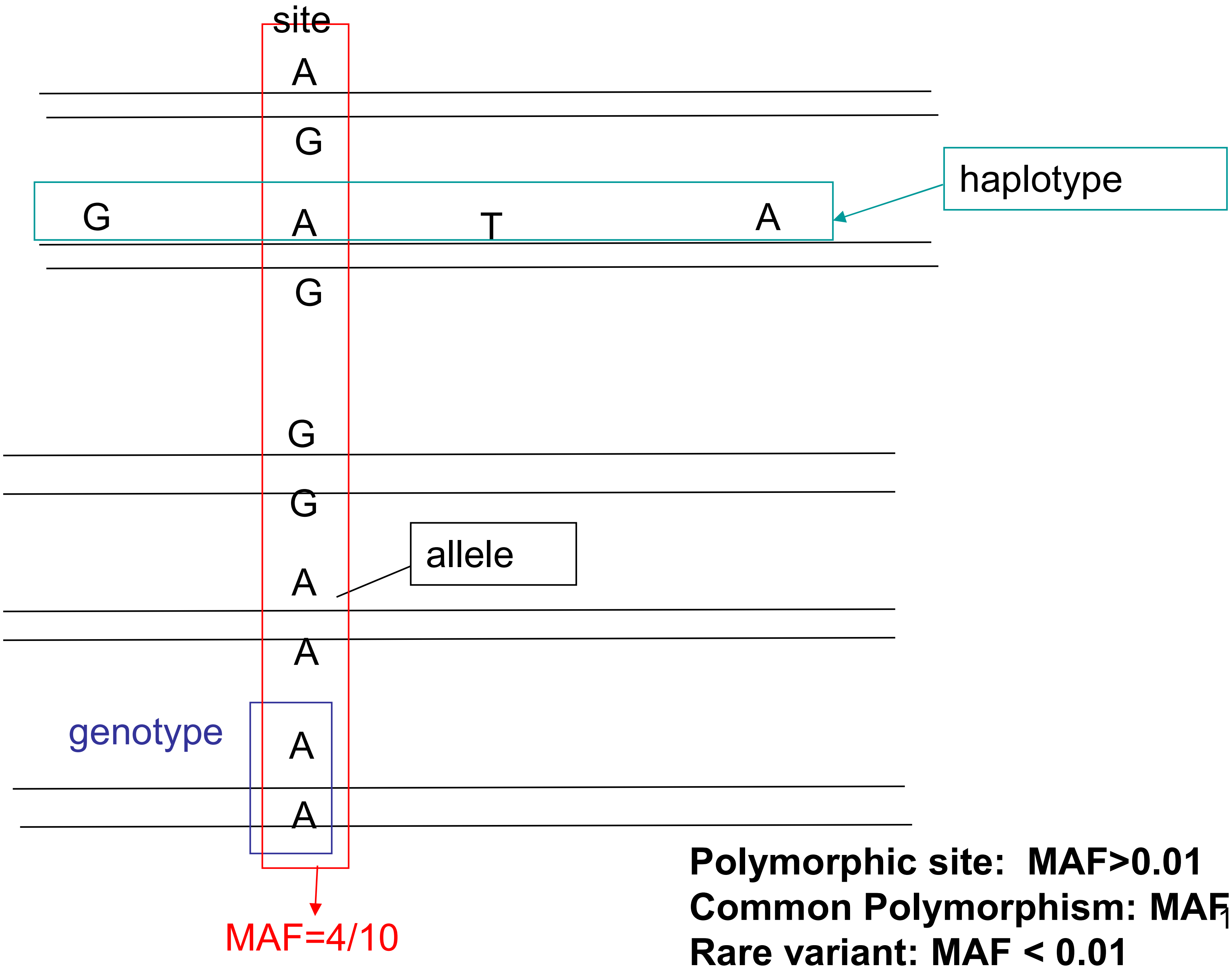
Fine-scale genetic variation between human populations is interesting as a signature of historical demographic events and because of its **potential for confounding disease studies**

The next step?

Resources to help to understand the genetic contribution to disease

1. ENCODE (2000-2002)
2. HapMap (2002-2007)
3. 1000 genomes (2008-2012)

Polymorphism & SNPs



Polymorphic site: MAF>0.01
Common Polymorphism: MAF≥0.05
Rare variant: MAF < 0.01

Polymorphism & SNPs

- **Allele:** One of several forms of a gene; at the DNA sequence level it refers to one of several (usually, 2) nucleotide sequences at a particular position in the genome.
- **Genotype:** The two specific alleles present in an individual; called a homozygote or heterozygote depending on whether the two alleles are identical or different.
- **Haplotype:** a collection of specific alleles along a chromosome
- **Polymorphism:** The occurrence of multiple alleles at a specific site in the DNA sequence.
- **MAF (minor allele frequency)** – an attribute of a site indicating the frequency of its minor allele in the population
- **Polymorphic site:** a site is called polymorphic if the rarer of the two alleles, called the minor allele, has a frequency above 0.01 in the population.
- **Common polymorphism**: MAF > 0.05.
- **Rare variant**: MAF< 0.01
- **SNP (single nucleotide polymorphism):** Polymorphism where multiple (usually, 2) bases (alleles) exist at a specific genomic sequence site within a population, such as A and G. In individuals, the possible combinations (genotypes) may be homozygous (AA or GG) or heterozygous (AG).

Facts that may help in association studies

ENCODE data: 10 regions, each of 500kb, sequenced in 269 DNA samples. (2000-2002)

Only 45% of SNPs are common. But -

1. In total, around 10M common polymorphic sites (every 300bp on average).

2. For any two homologous chromosomes, 1:1000 bp are heterozygous SNPs. In 90% of cases, the SNP is common.

The common disease- common variant hypothesis (CD/CV)

The CD/CV hypothesis: Common variants could have a causal role in common diseases.

Rests on the following promise:

→ For common disease, many susceptibility traits may have been only mildly deleterious or neutral (e.g., late onset diseases).

← Because the vast majority of variation in the population is due to common variants, the susceptibility alleles for a trait will include many common variants.

Systematic testing of association for common human diseases

The needs:

1. A data of sequence variation (for 10 million common SNPs?)
2. Laboratory tools to identify variation in large scale
3. Computational methods to interpret data

A considerable controversy (2000-2005)

© 2002 Oxford University Press Human Molecular Genetics, 2002, Vol. 11, No. 20 2417–2423

The allelic architecture of human disease genes: common disease–common variant . . . or not?

Jonathan K. Pritchard* and Nancy J. Cox

Department of Human Genetics, University of Chicago, 920 E 58th St–CLSC 507, Chicago IL 60637, USA

Molecular Psychiatry (2004) 9, 227–236
© 2004 Nature Publishing Group All rights reserved 1359-4184/04 $25.00
www.nature.com/mp

FEATURE REVIEW

Will haplotype maps be useful for finding genes?

EJCG van den Oord[1] and BM Neale[1]

Virginia Institute for Psychiatric and Behavioral (
Richmond, VA, USA

News focus

Hapmap flap

Mediawatch: **Richard F. Harris** contrasts the media's response two years ago to the announcement of the sequencing of the human genome sequence with the new haplotype mapping project to determine genetic links to common diseases.

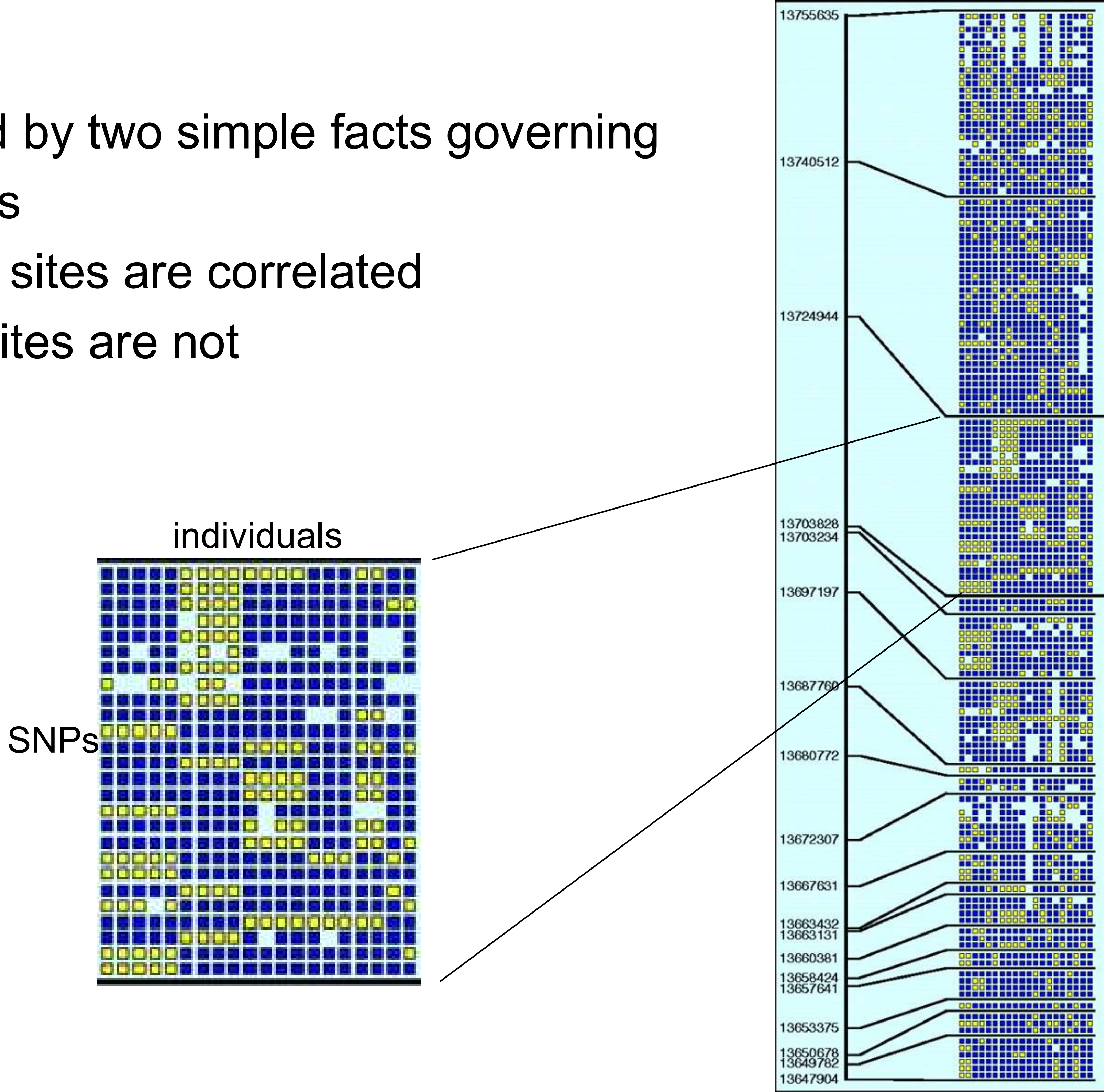
executive of Decode Genetics of Reykjavik, Iceland, said he thought the hapmap 'would be an inefficient way of finding disease genes.'

How many SNPs do we really need?

Does the genome suggest potential shortcuts?

We are guided by two simple facts governing polymorphisms

1. Nearby sites are correlated
2. Distal sites are not



ENCODE data

Linkage disequilibrium (LD):

The statistical correlation between alleles at two polymorphic sites along the genome in a population

Linkage disequilibrium

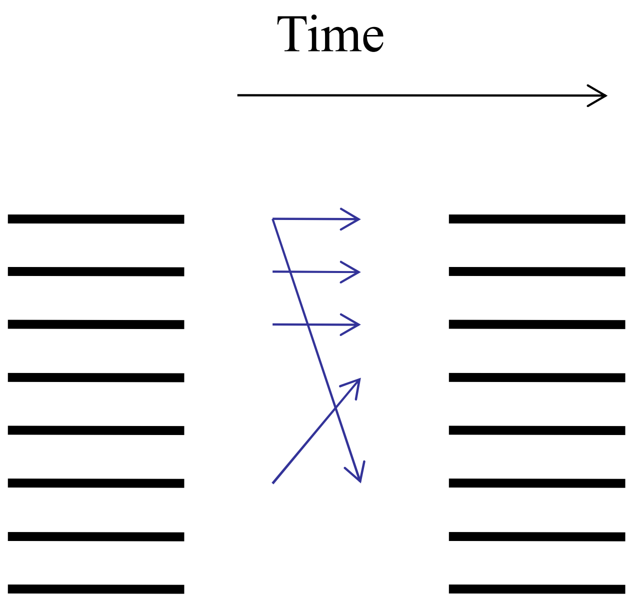
We are guided by two simple facts governing polymorphisms

1. Nearby sites are correlated
2. Distal sites are not

1. The genetics: why should this happen?
2. The visualization
3. The organization in blocks and reduced haplotype complexity
3. The computation – Linkage Disequilibrium
4. The utility of LD in GWAS

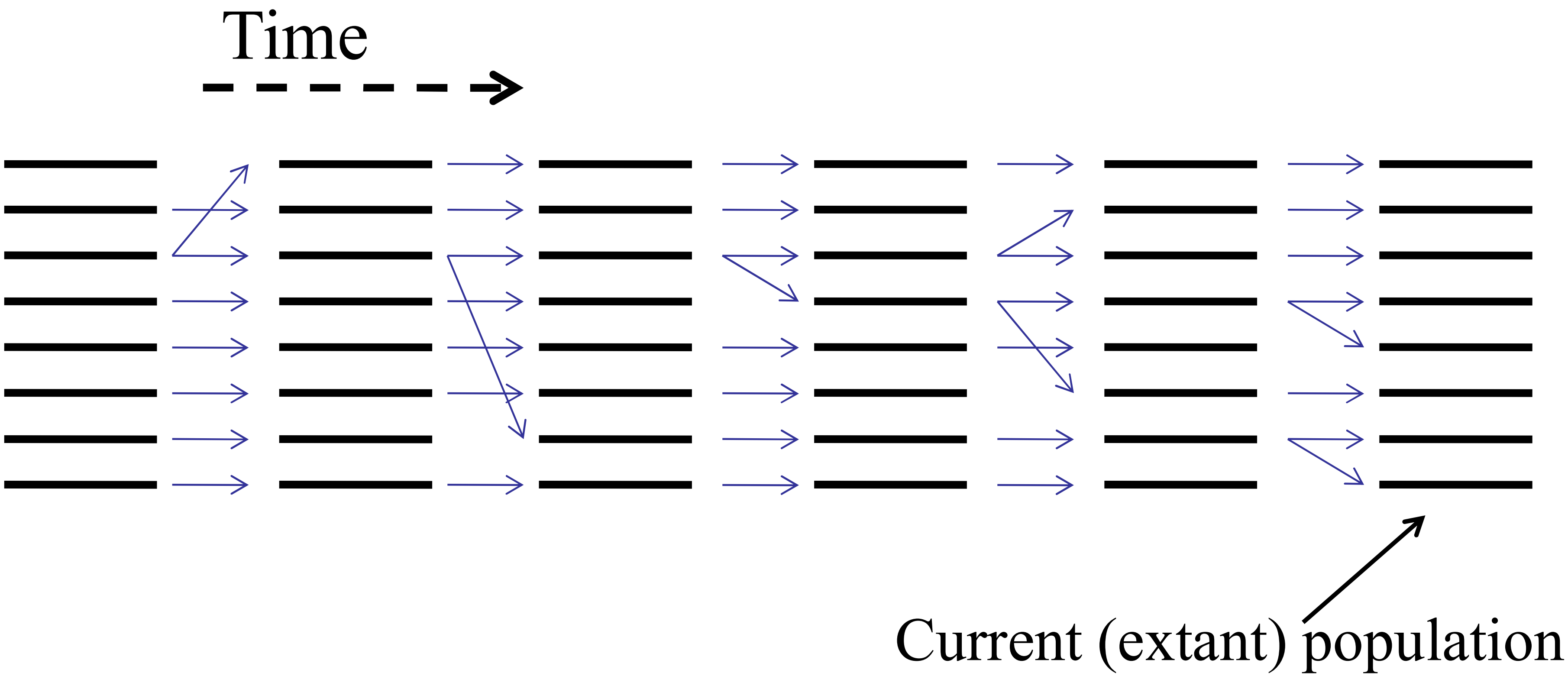
So, why should the proximal SNPs be correlated, and distal SNPs not?

A bit of evolution

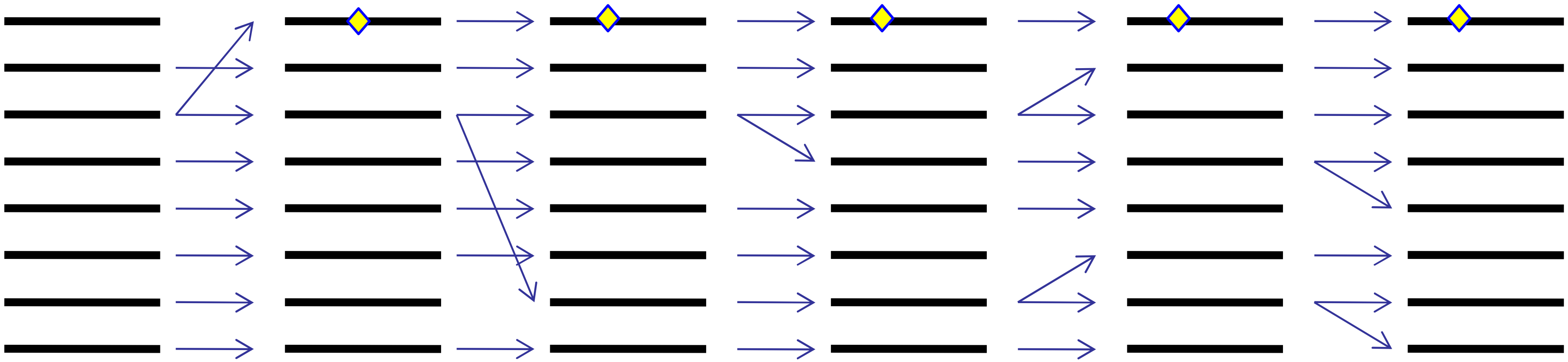


- Consider a fixed population (of chromosomes) evolving in time.
- Each individual arises from a unique, randomly chosen parent from the previous generation

Genealogy of a chromosomal population

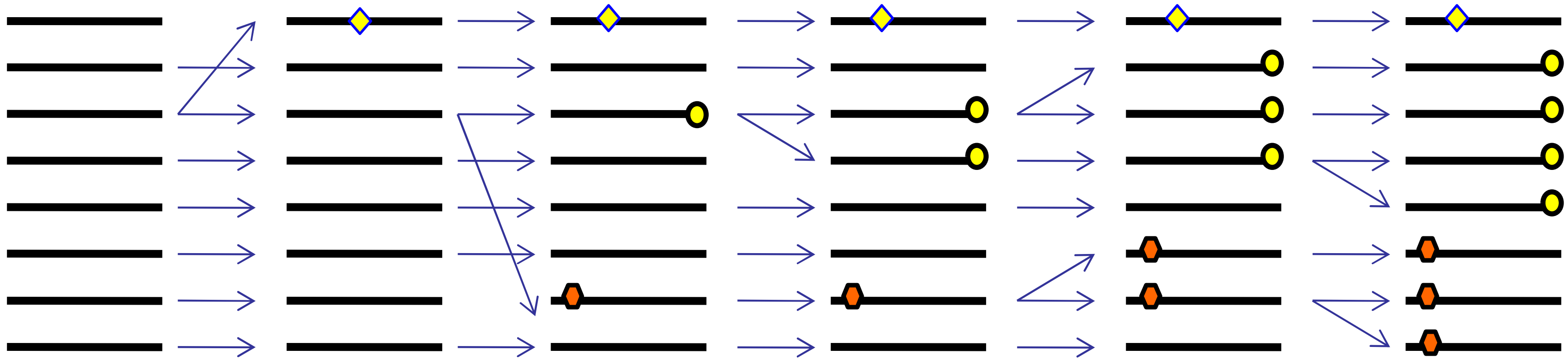


Adding mutations



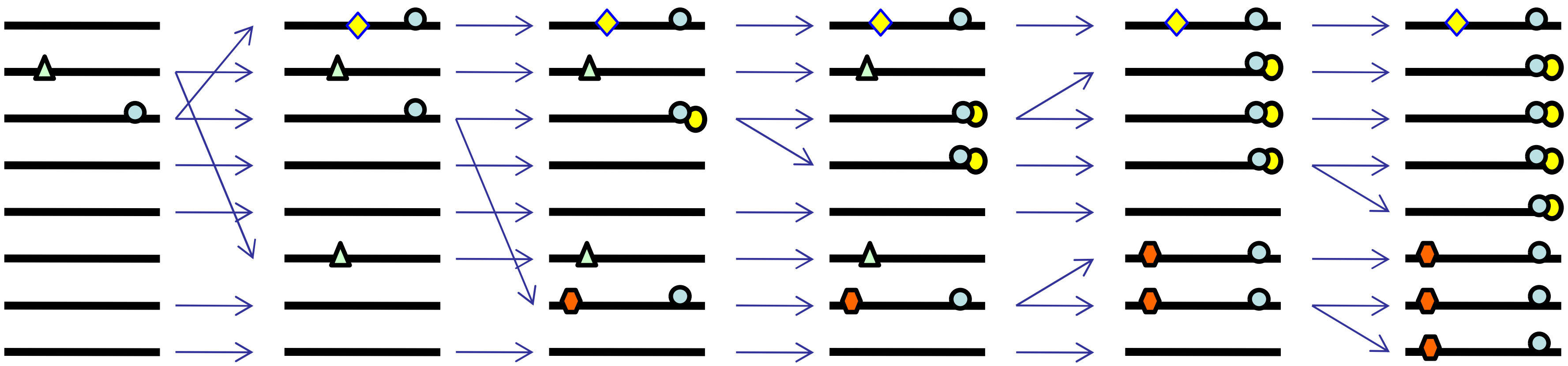
Infinite sites assumption: A mutation occurs at most once at a site.

SNPs

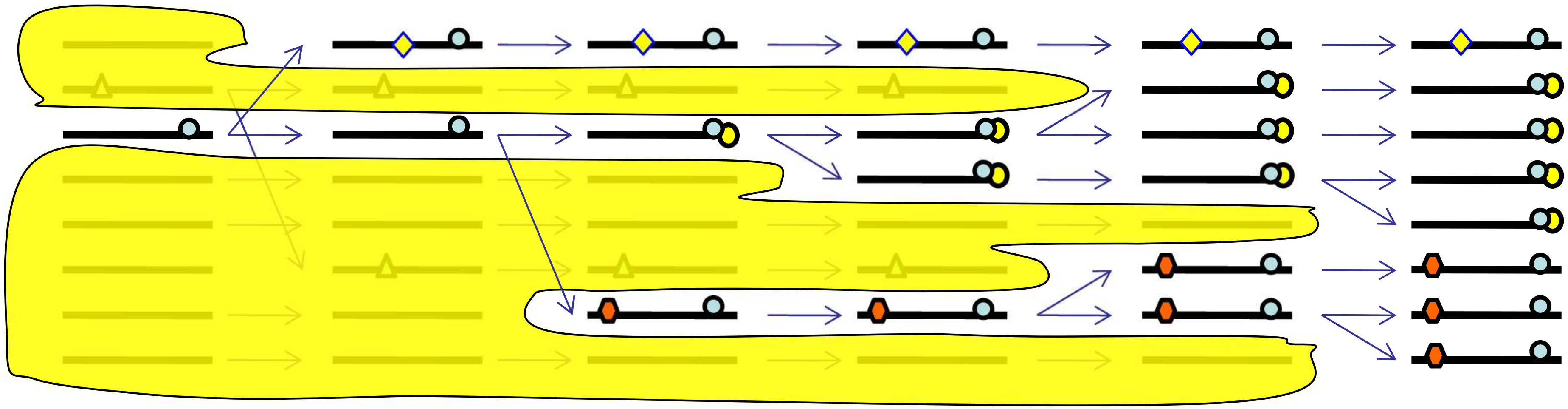


The collection of acquired mutations in the extant population describe the SNPs

Fixation and elimination

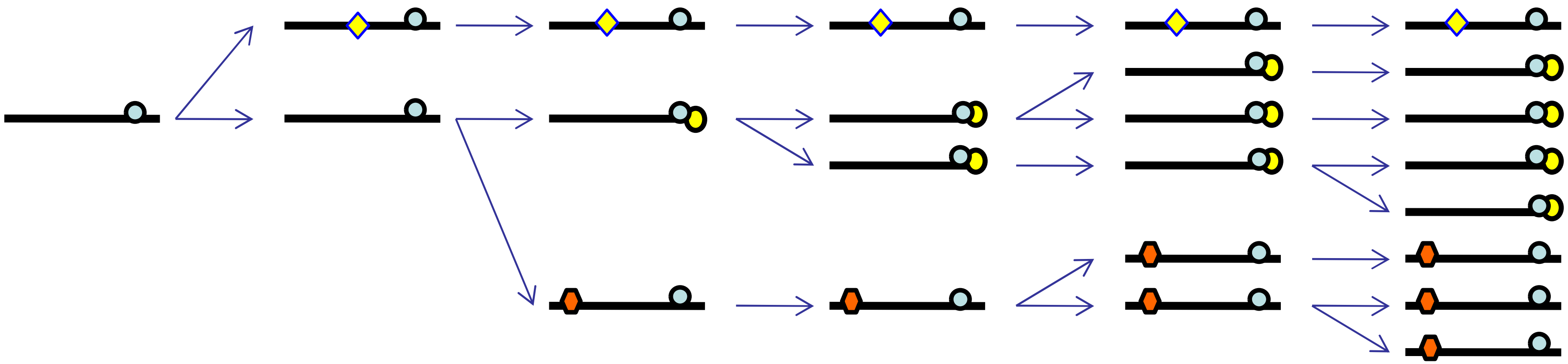


- Not all mutations survive.
- Some mutations get fixed, and are no longer polymorphic

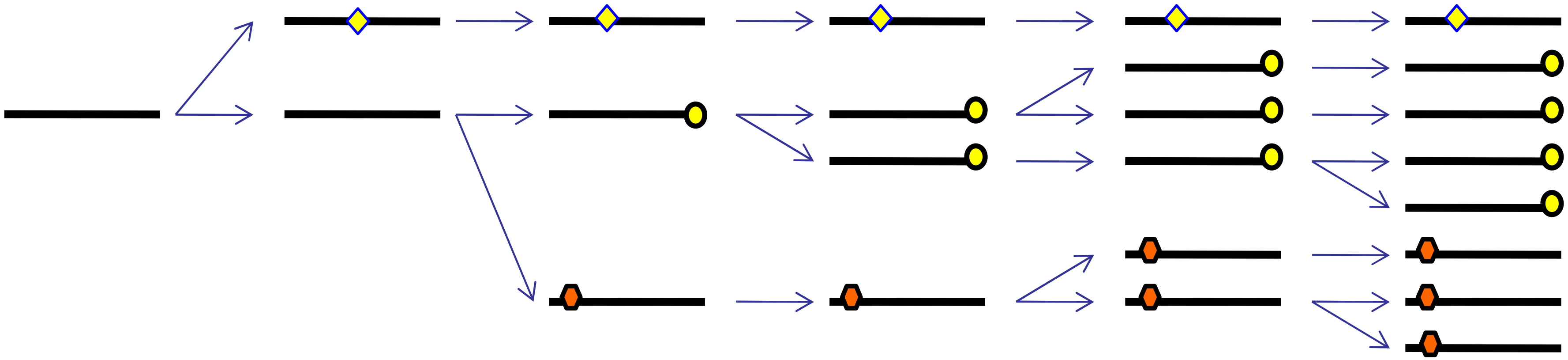


Removing extinct genealogies

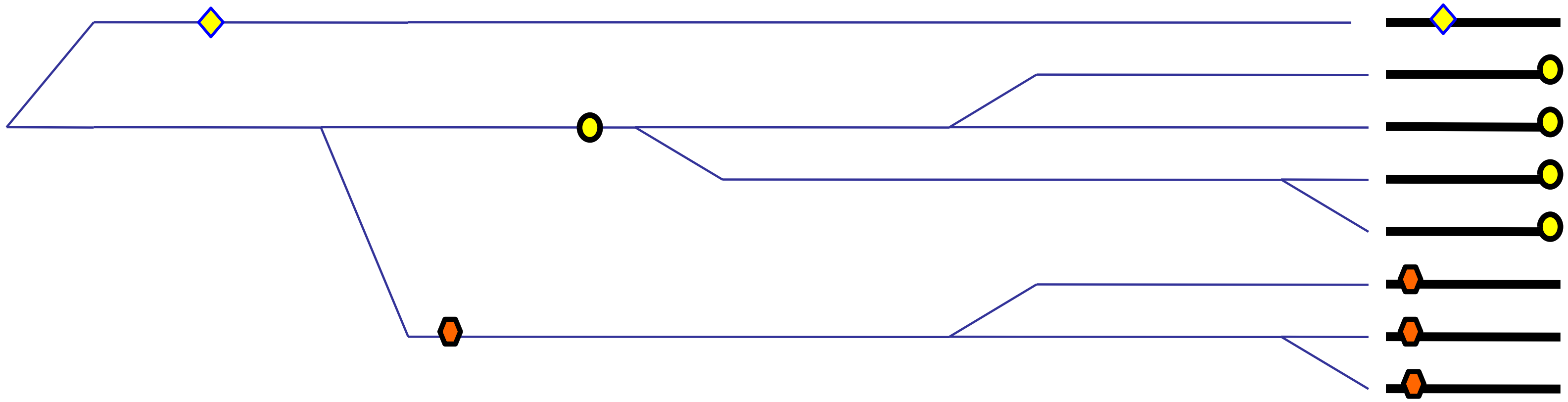
Removing fixed mutations



The coalescent

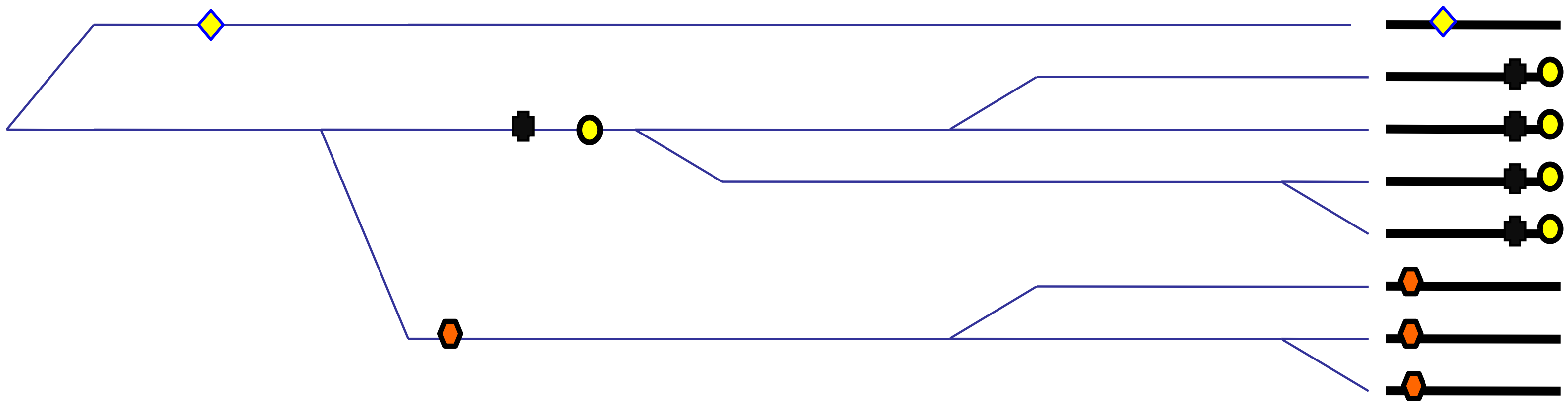


The coalescent



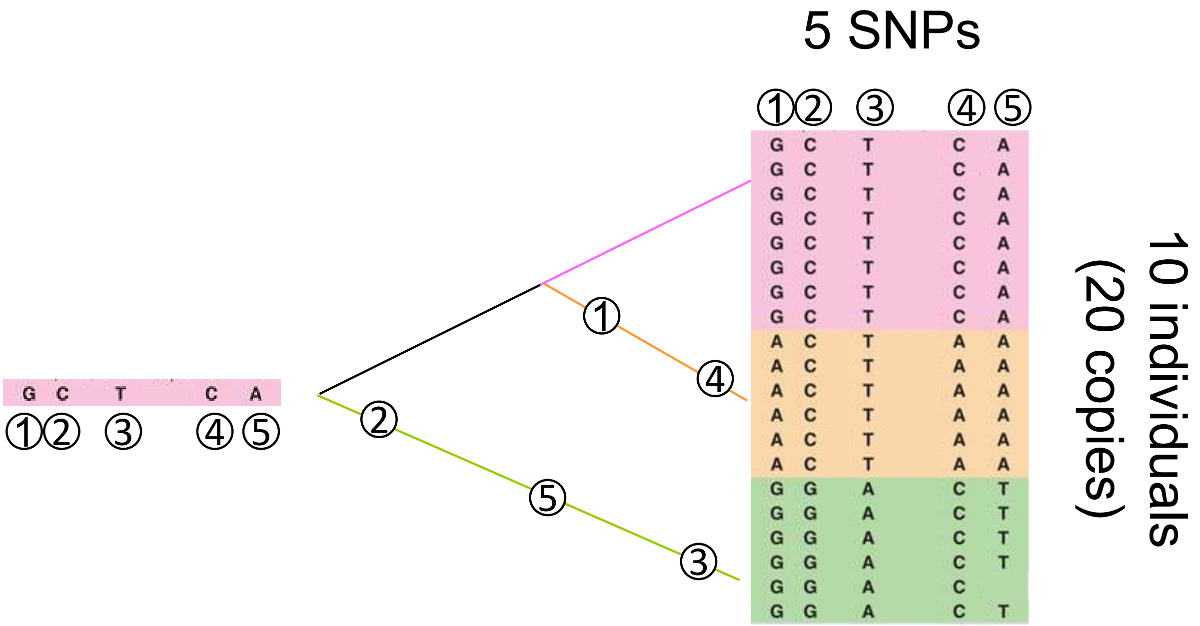
- We drop the ancestral chromosomes, and place the mutations on the internal branches.

The coalescent



- Note that the tree (genealogy) is hidden.
- However, the underlying tree topology introduces a correlation between each pair of SNPs

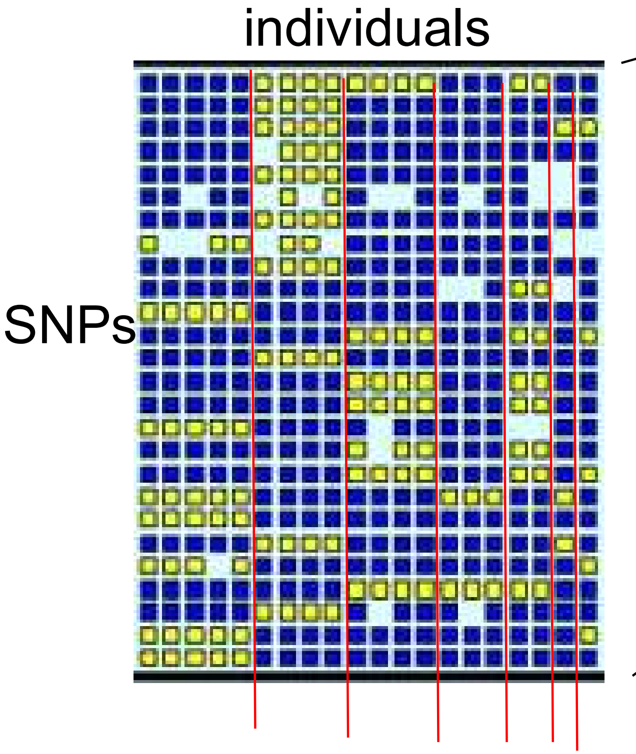
The coalescent

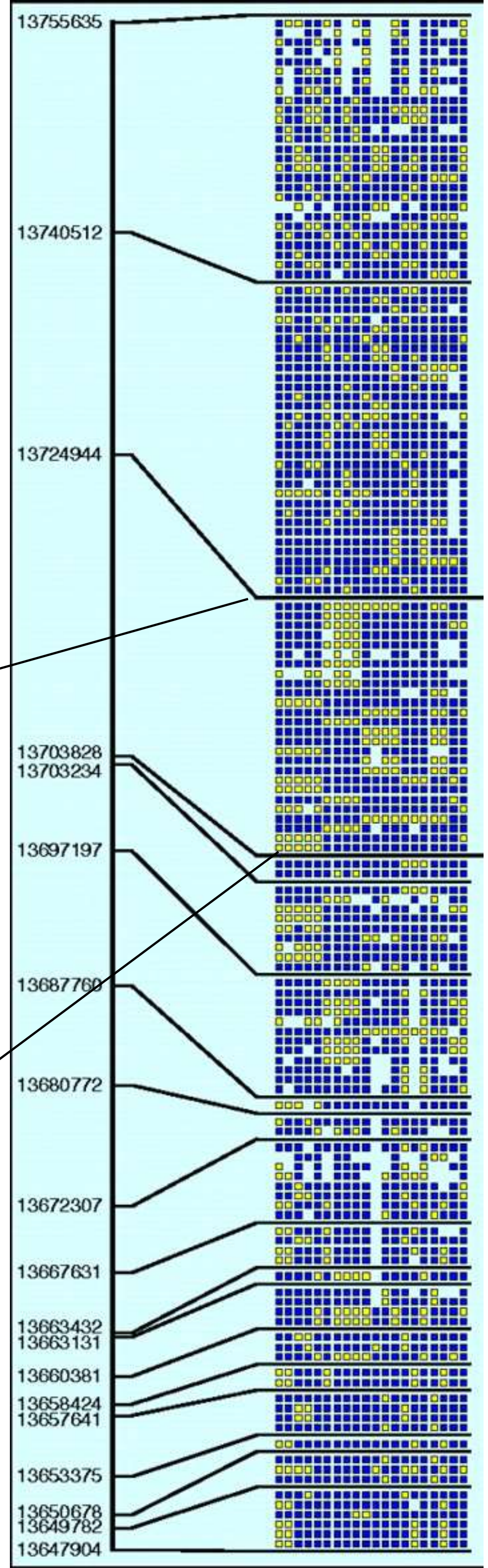


Only 3 haplotypes!

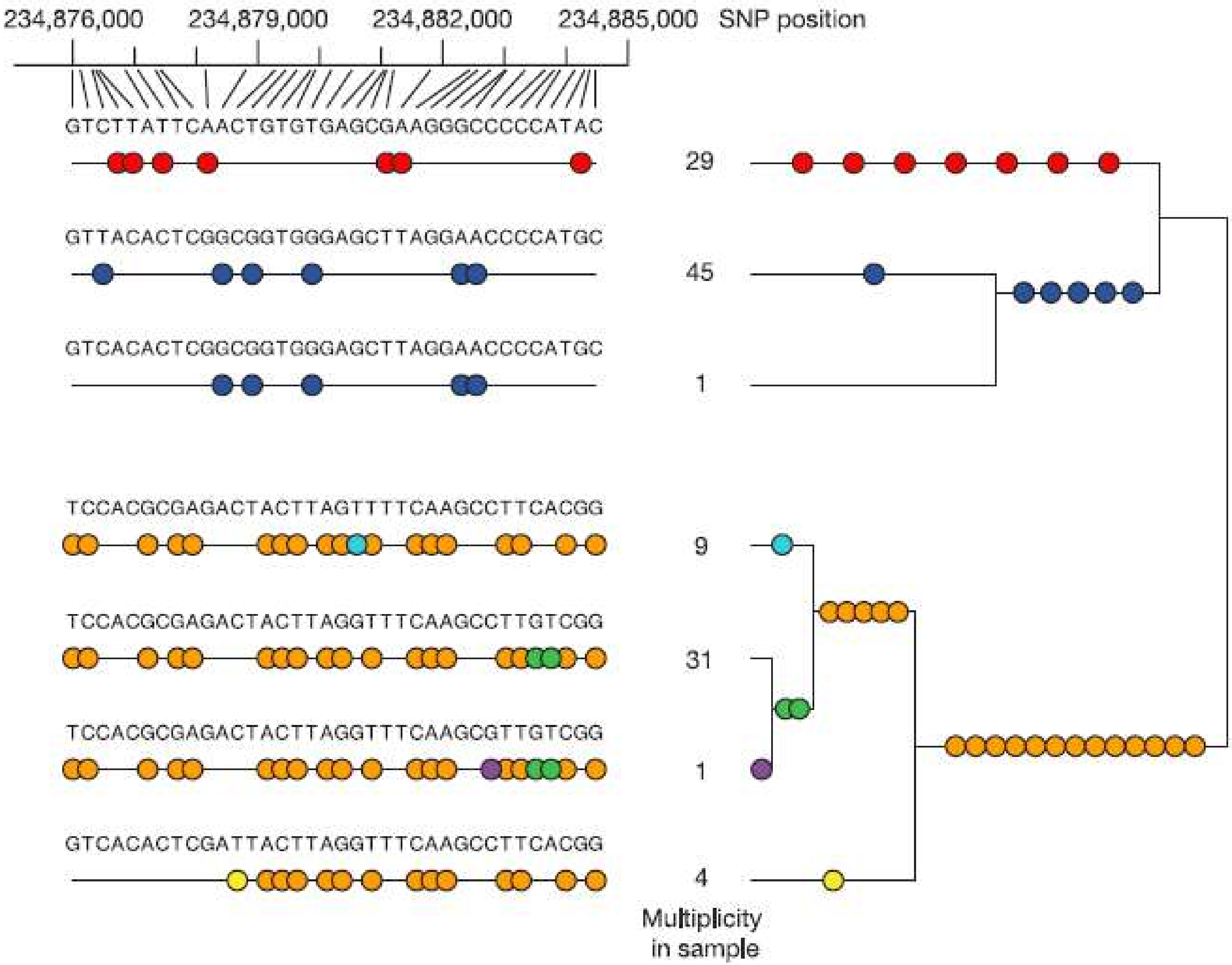
Altshuler et al.
Science 2008

Example: 26 SNPs and only 7 different haplotypes in human chromosome 21



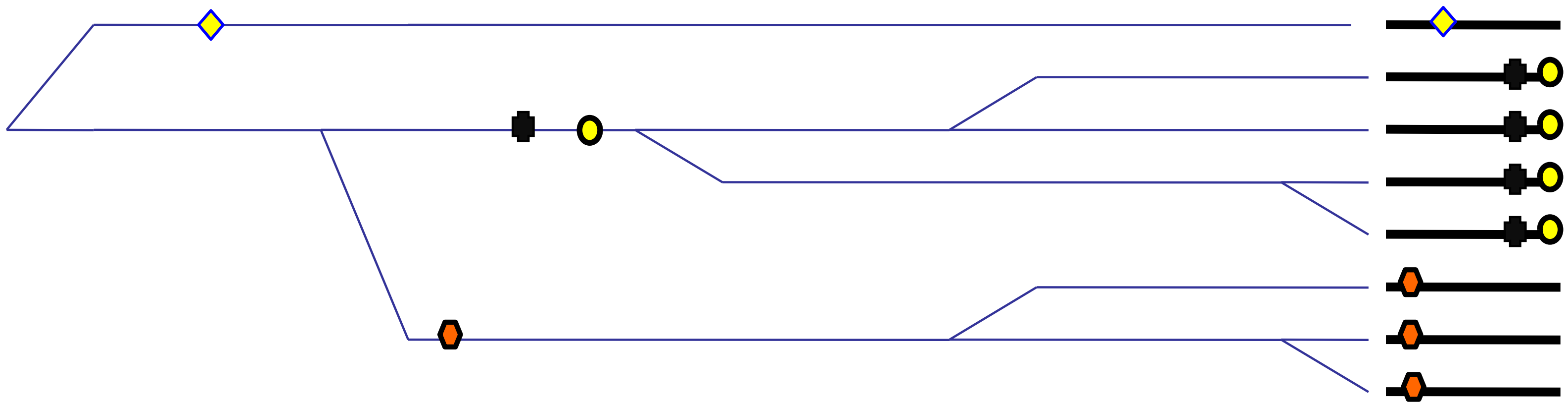


Example: 36 SNPs and only 7 different haplotypes in chromosome 2



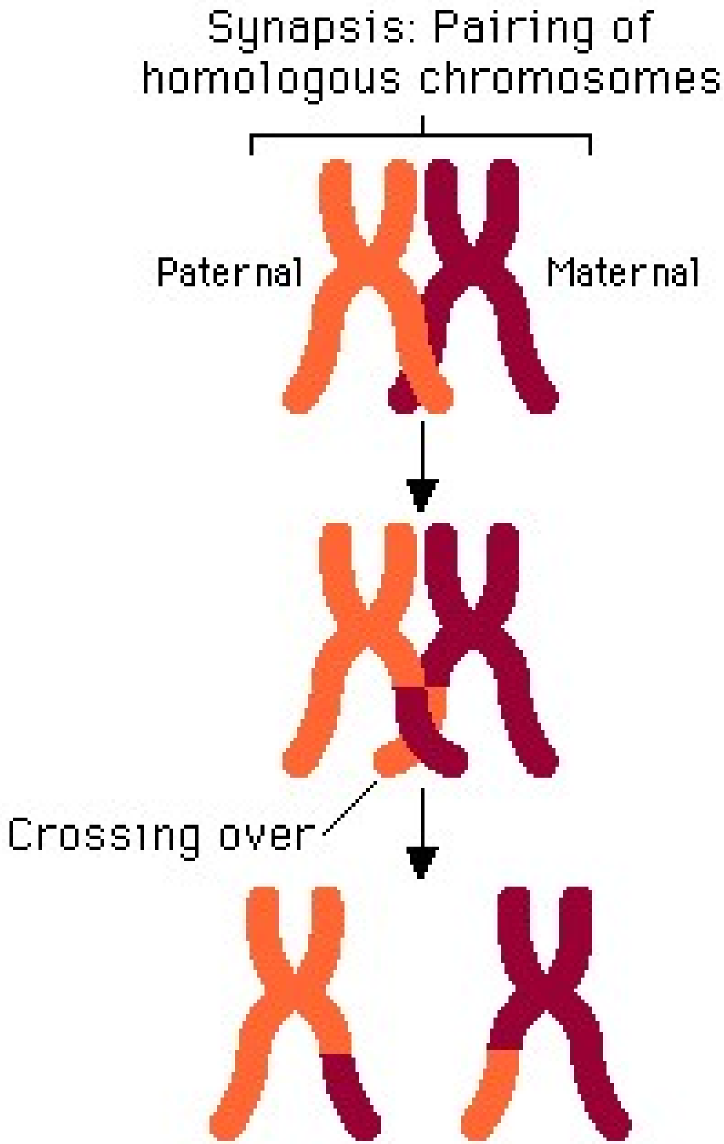
The region of chromosome 2 (234,876,004–234,884,481 bp; NCBI build 34) within ENr131.2q37 contains 36 SNPs, with zero obligate recombination events in the CEU samples.

What have we learnt?

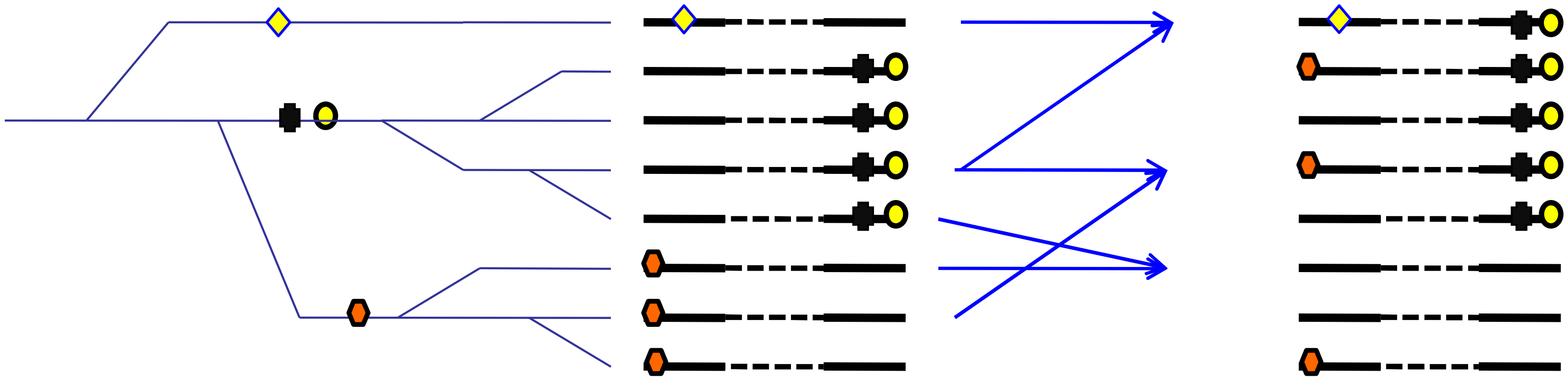


- The underlying genealogy creates a correlation between SNPs.
- By itself, this is not sufficient, because distal SNPs might also be correlated.
- Fortunately, for us the correlation between distal SNPs is quickly destroyed.

Recombination

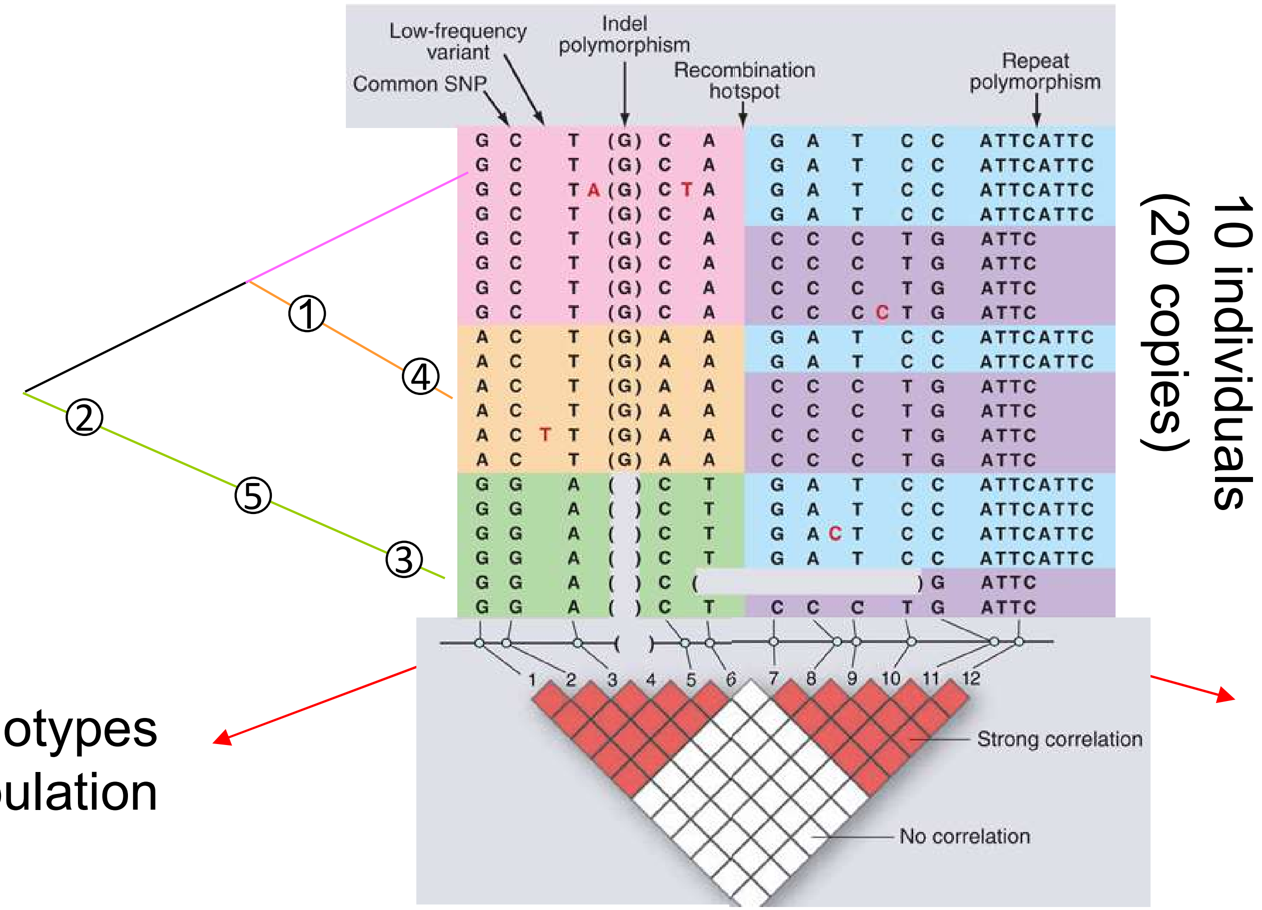


Multiple recombination change the local genealogy



- Proximal SNPs are correlated, distal SNPs are not.

Variation in recombination rate is a major determinant of LD blocks

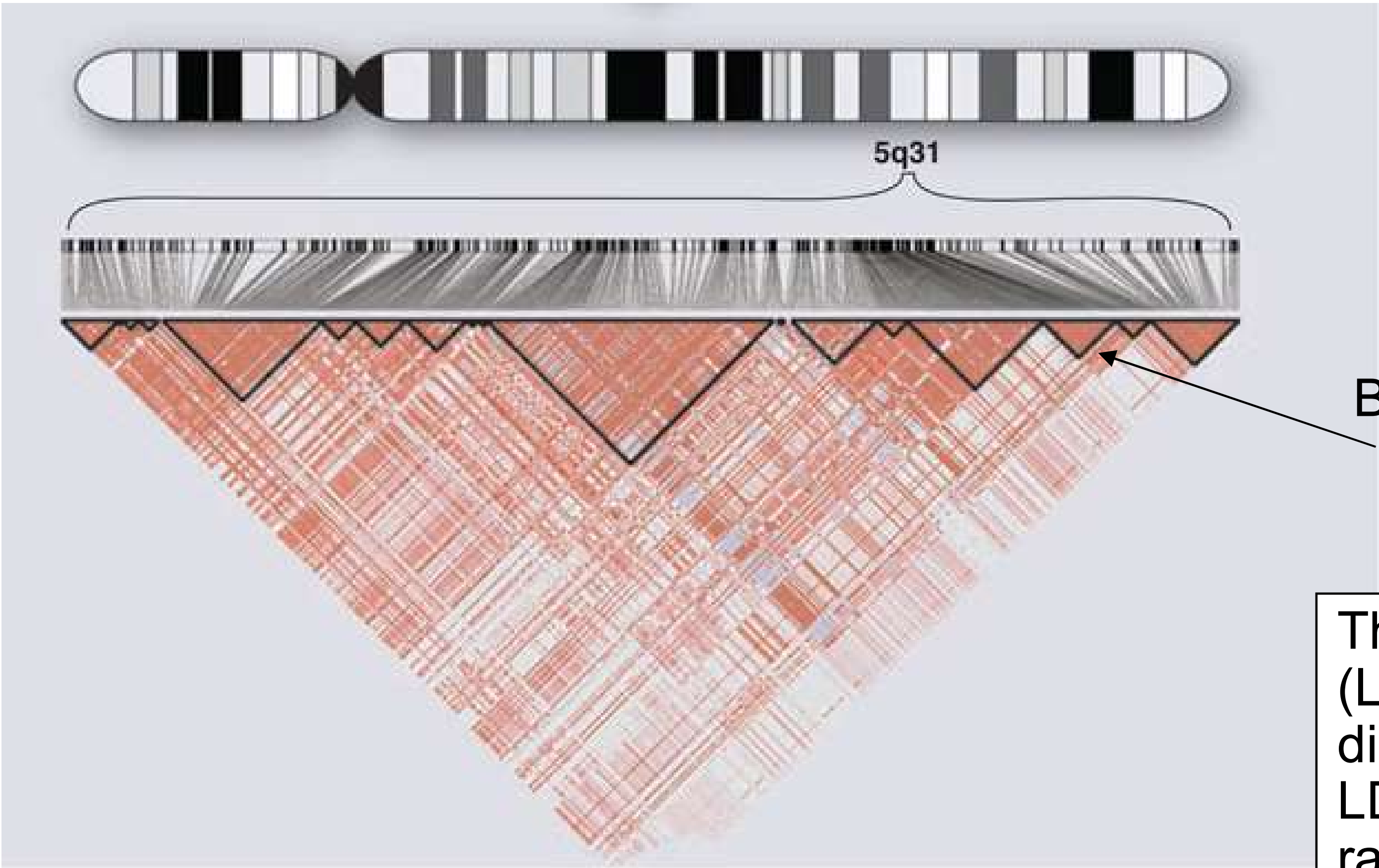


3 haplotypes in population

6 haplotypes in population

Altshuler et al. Science 2008

A block-like structure of human LD is visually apparent



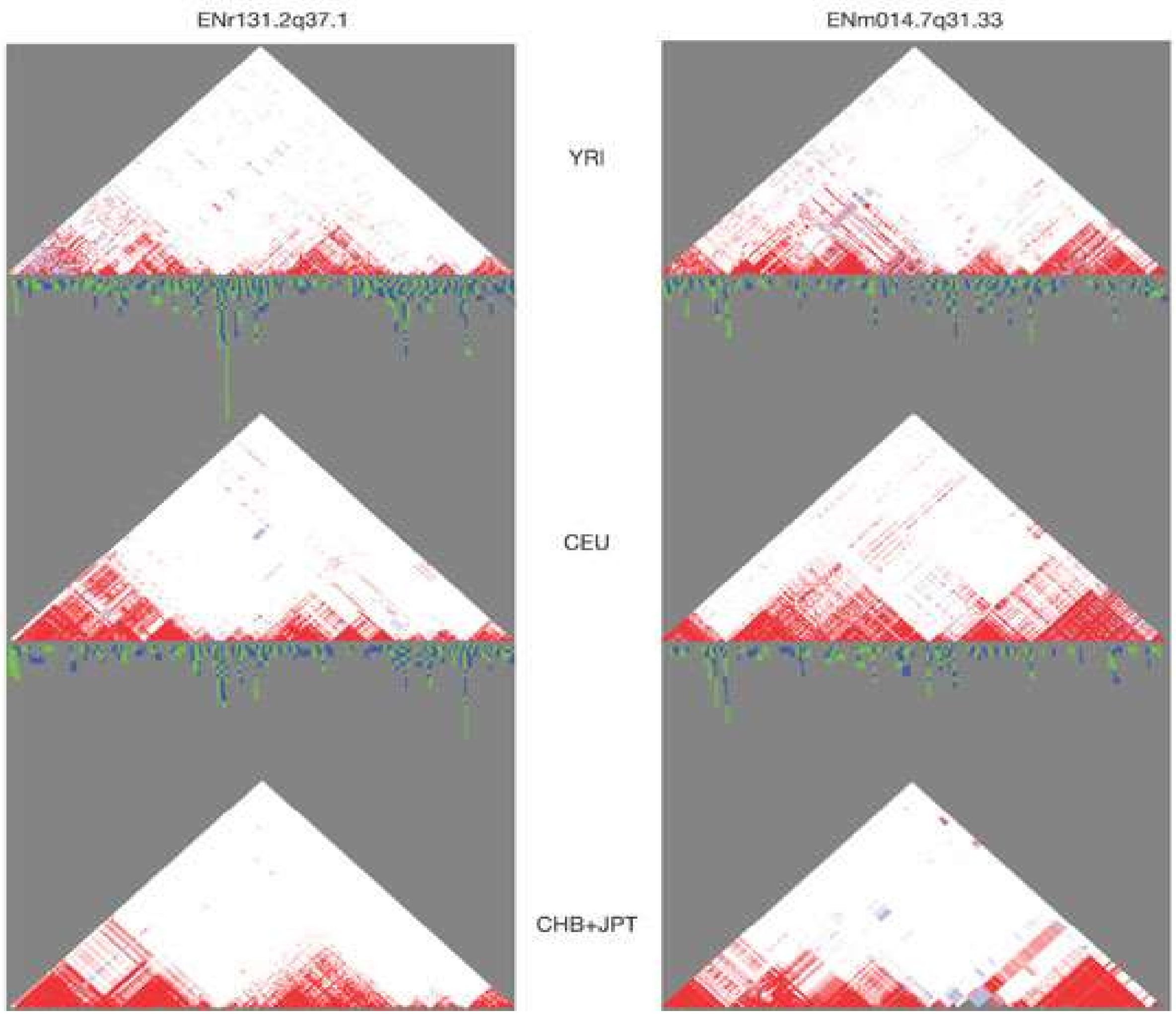
Blocks of strong correlation

The correlation (Linkage disequilibirium, LD) decays rapidly after 20-50kb

- Data from the International HapMap Project
- Showing 420 genetic variants in a region of 500 kb on human chromosome 5q31.
- Longer-range patterns are often complex because weaker recombination hotspots may reduce, but not completely eliminate, marker-to-marker correlation.

Altshuler et al. Science 2008

Comparison of LD in 3 populations



YRI - 90 individuals from the Yoruba in Ibadan, Nigeria
CEU - 90 individuals in Utah, USA.
CHB - 45 Han Chinese in Beijing, China
JPT – 44 Japanese in Tokyo, Japan

Genetic structure of human variation

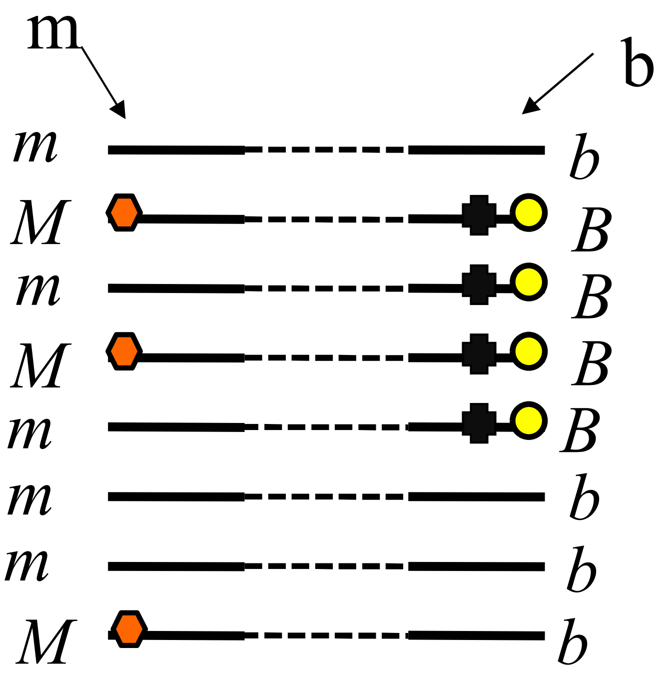
1. Extensive linkage disequilibrium: Variation occurs in segments separated by sites of recombination.
2. The vast majority of heterozygous sites in each person are in LD with a limited set of common SNPs that were contained in dbSNP in 2002.

Statistical tests of correlation (linkage disequilibrium, LD) among SNPs

- **Several metrics have been devised to measure linkage disequilibrium (LD).**
- **Commonly used metrics: D,*D′*, r^2 , chi-sq, p-values**

Assigning confidence: $\chi 2$ test

Step 1: Fill-in observed 2x2 matrix

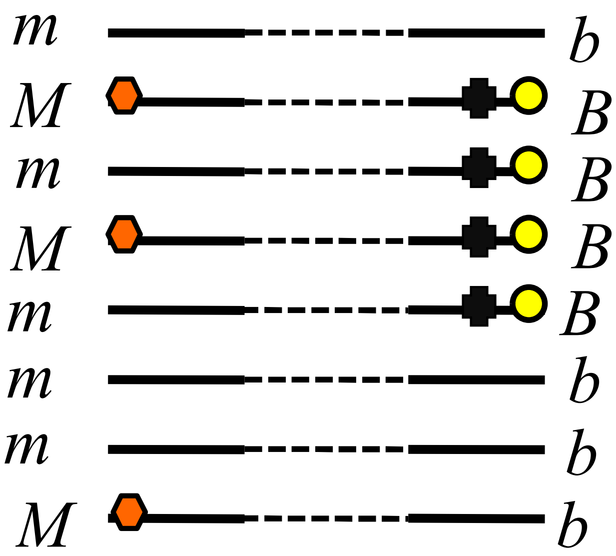


Observed

	3	**2**
	1	2

Assigning confidence: $\chi 2$ test

Step 2: calculate probabilities



Observed

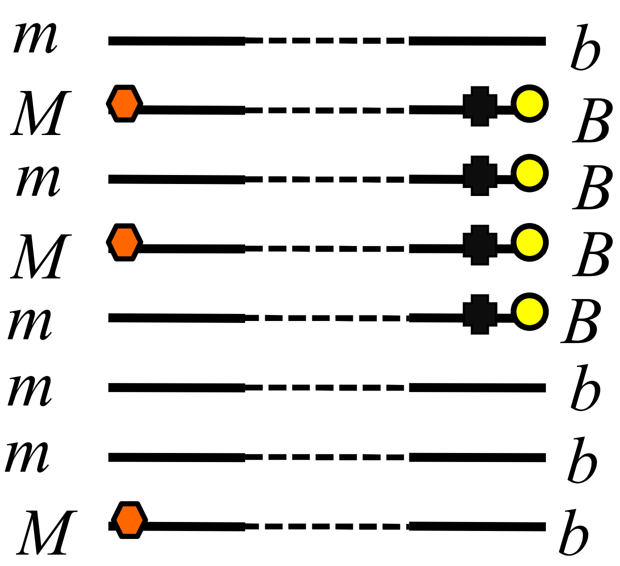
	3	**2**
	1	2

	b	B	
m	**3**	**2**	$p(m) = 5/8$
M	1	2	$p(M) = 3/8$
	$p(b) = 4/8$	$p(B) = 4/8$	

n=8

Assigning confidence: $\chi 2$ test

Step 3: Fill-in Expected 2x2 matrix



Expected

	b	B	
m	$p(m)p(b)n$	$p(m)p(B)n$	$\rightarrow p(m)$
M	$p(M)p(b)n$	$p(M)p(B)n$	$\rightarrow p(M)$
	$\downarrow p(b)$	$\downarrow p(B)$	

Expected

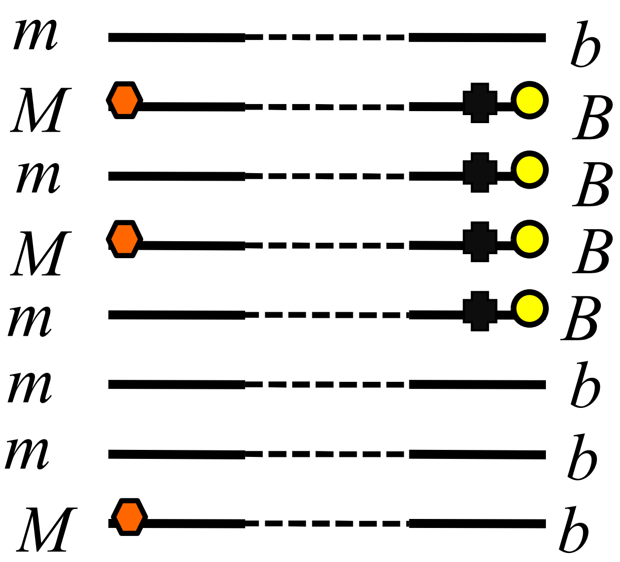
	b	B	
m	**5/8*4/8*8**	**5/8*4/8*8**	$\rightarrow p(m) = 5/8$
M	3/8*4/8*8	3/8*4/8*8	$\rightarrow p(M) = 3/8$
	$\downarrow p(b) = 4/8$	$\downarrow p(B) = 4/8$	

Expected

	b	B
m	**2.5**	**2.5**
M	1.5	1.5

Assigning confidence: $\chi 2$ test

Step 3: Calculate $\chi 2$



Expected

	2.5	**2.5**
	1.5	1.5

E_1	E_2
E_3	E_4

Observed

	3	**2**
	1	2

O_1	O_2
O_3	O_4

$$\chi^2 = \sum_i \frac{(O_i - E_i)^2}{E_i} = \left(\frac{0.25}{2.5} + \frac{0.25}{2.5} + \frac{0.25}{1.5} + \frac{0.25}{1.5} \right) = 0.53$$

The utility of LD in association studies

The utility of LD in association studies

1. Association test based on a proxy SNP

2. Selection of tag SNPs for association test

1. Association test based on a proxy SNP

$$\mathbb{E}[\chi^2] \propto N\gamma^2 p(1-p)r^2,$$

where χ^2 is the chi-squared test statistic, N the number of cases and controls, γ the effect size, p the allele frequency of the risk variant and r^2 is the correlation between the marker and causal SNP.

gamma=Effect size = Genetic odds ratio

The higher correlation r^2, the lower the sample size required to maintain the same power

Epidemiological term: Odds ratio

- Odds – The ratio of the probability of an event's occurring to the probability of its not occurring
- The odds ratio (OR) is a way of comparing whether the odds of a certain event is the same for two groups.

	Group 0	Group 1
Event a	r_0	r_1
Event b	s_0	s_1
	Odds = r0/s0	Odds = r1/s1

$$OR = \frac{r1/s1}{r0/s0}$$

Genetic odds ratio

- Groups = genotypes,
- an event = case (disease)

	AA	AB
Case	r_0	r_1
Control	s_0	s_1

$$OR = \frac{r1/s1}{r0/s0}$$

Epidemiological term - Relative risk

- Risk = the probability of cases in a group.
- Relative risk (RR) is a way of comparing whether the risk is the same for two groups.

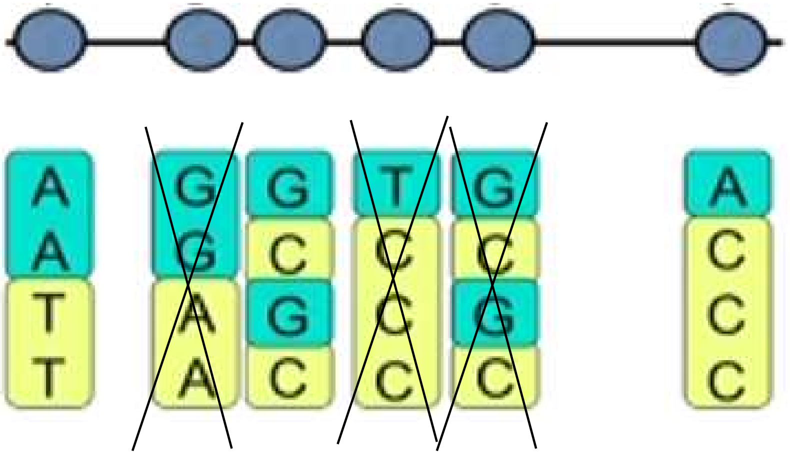
	AA	AB
Case	r_0	r_1
Control	s_0	s_1

f_0= Risk = r0/(r0+s0)

f_1= Risk = r1/(r1+s1)

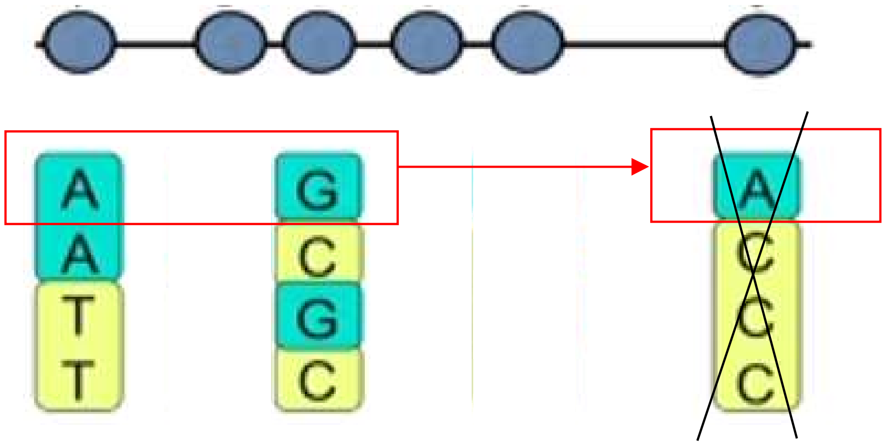
$$RR = \frac{r1/(r1+s1)}{r0/(r0+s0)}$$

2. Selection of tag SNPs for association test



Step 1: tagging using a simple pairwise method. SNPs are selected for genotyping until all common SNPs are highly correlated ($r^2 > 0.8$) to one or more members of the tag set. (may prioritize tags based on the number of other SNPs captured).

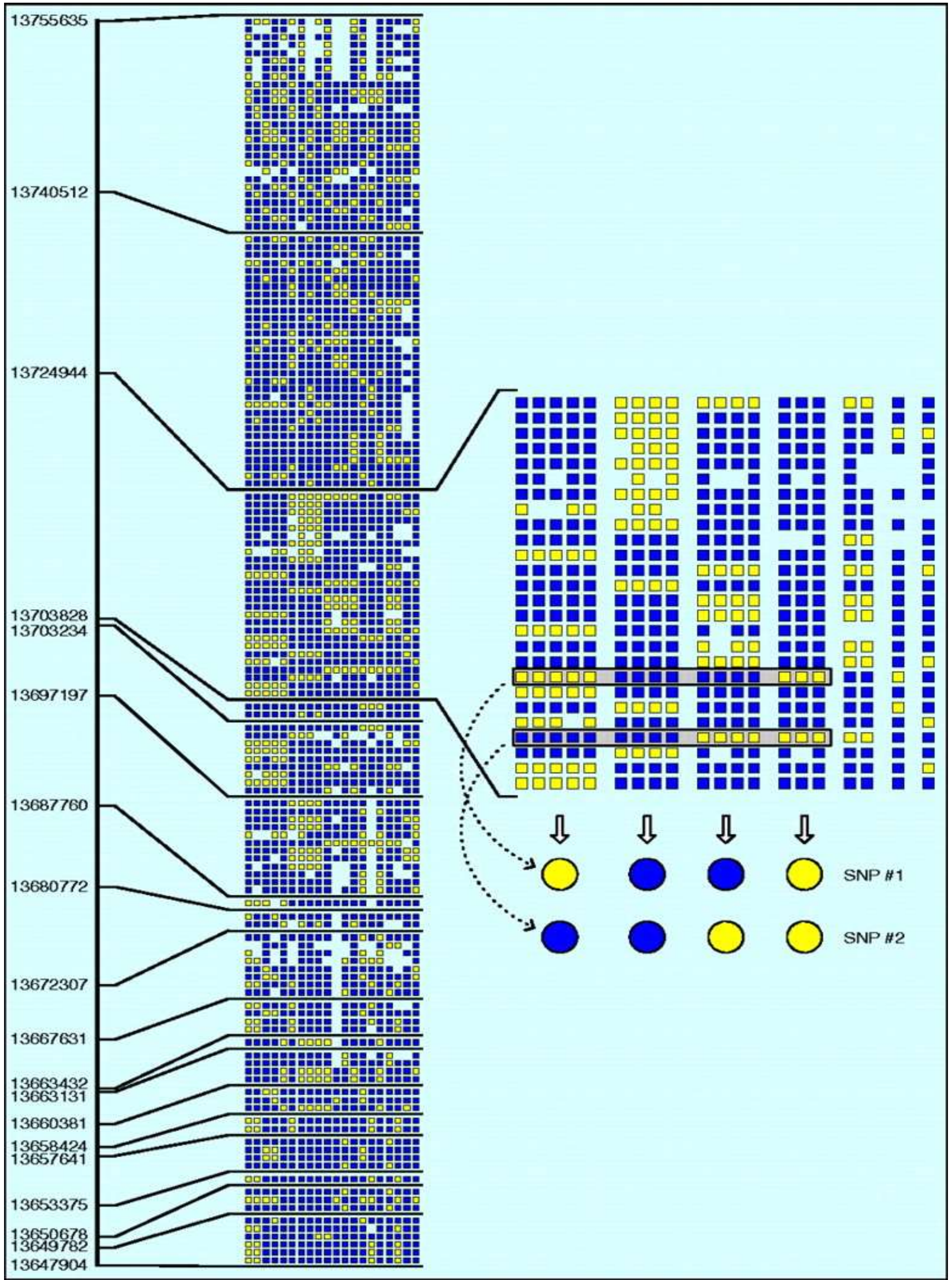
2. Selection of tag SNPs for association test



Step 1: tagging using a simple pairwise method. SNPs are selected for genotyping until all common SNPs are highly correlated ($r^2 > 0.8$) to one or more members of the tag set. (may prioritize tags based on the number of other SNPs captured).

Step 2: improve through exploiting multimarker haplotypes.

2. Selection of tag SNPs for association test



The HapMap project

HapMap Samples

- 90 Yoruba individuals (30 parent-parent-offspring trios) from Ibadan, Nigeria (YRI)
- 90 individuals (30 trios) of European descent from Utah (CEU)
- 45 Han Chinese individuals from Beijing (CHB)
- 45 Japanese individuals from Tokyo (JPT)

@Daly, M

HapMap phase I

Project launched – October 2001

Phase I:

• Production 2002-2004.

• Completed October 2005.

• 1,000,000 SNPs typed in all 270 HapMap samples (from dbSNP)

• ENCODE variation reference resource available

The HapMap rationale

~1M common SNPs from dbSNP
MAF > 0.05 (phase I)

Genotyping of individuals

↓ Predict haplotypes ('phasing')

Select tag SNPs

100K/500K SNPs
(phase I)

↓

Produce genotyping array platforms (affymetrix, Illumina)

↓

Apply genome-wide association studies for any disease and any causal mutation

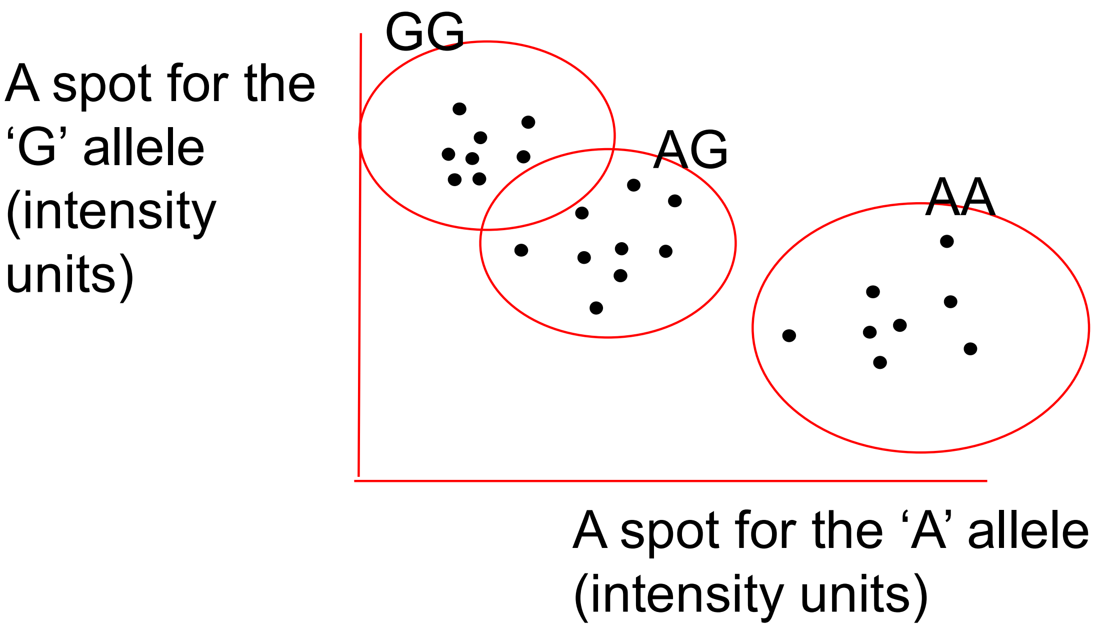
Estimated coverage of commercially available genotyping platforms

Table 1
Estimated coverage of commercially available fixed marker genotyping platforms

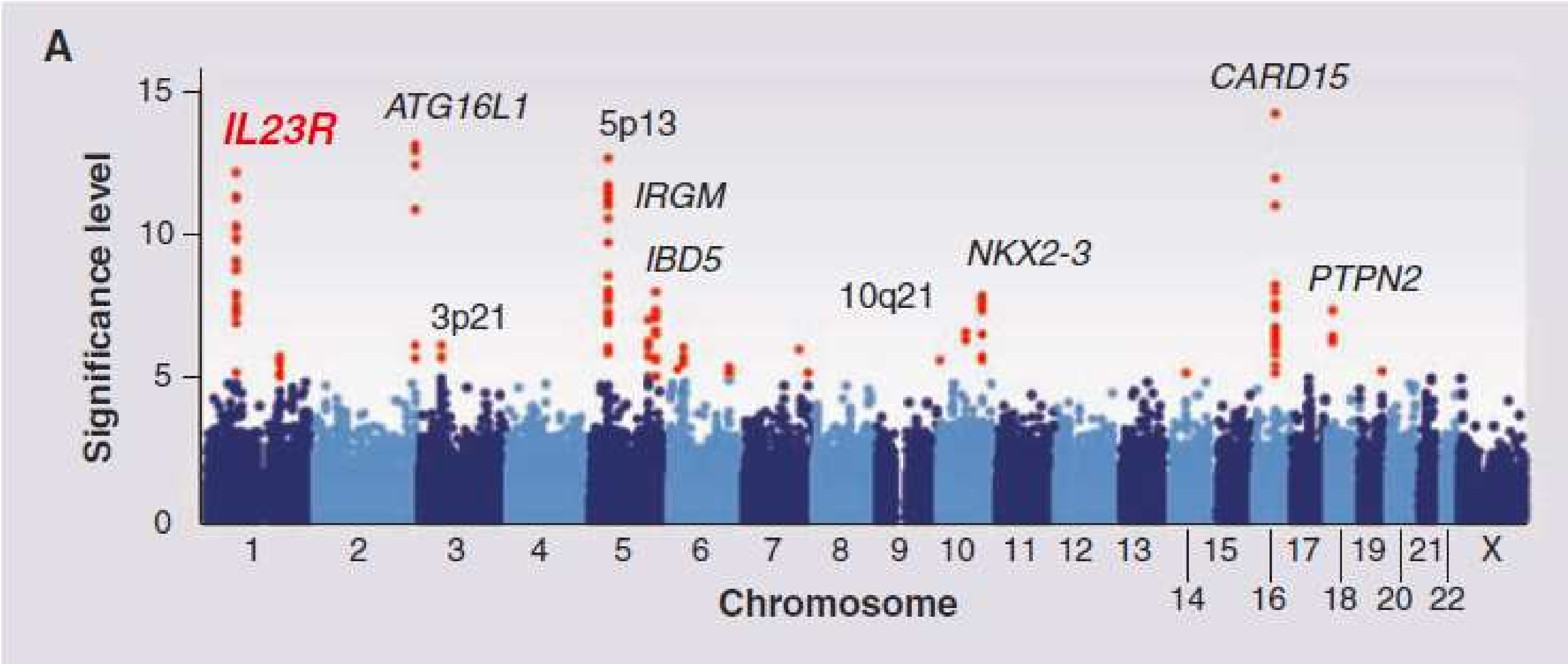
Platform	HapMap population sample YRI	CEU	CHB + JPT
Affymetrix GeneChip 500K	46	68	67
Affymetrix SNP Array 6.0	66	82	81
Illumina HumanHap300	33	77	63
Illumina HumanHap550	55	88	83
Illumina HumanHap650Y	66	89	84
Perlegen 600K	47	92	84

Data represent percent of SNPs tagged at $r^2 \geq 0.8$. Values assume all SNPs on the platform are informative and pass quality control. YRI, Yoruba in Ibadan, Nigeria; CEU, subsample of Utah residents of Northern European ancestry selected from Centre d'Étude du Polymorphisme Humain samples; CHB, Han Chinese in Beijing, China; JPT, Japanese in Tokyo. From the International HapMap Consortium, 2007 (3).

Reading genotype from an array



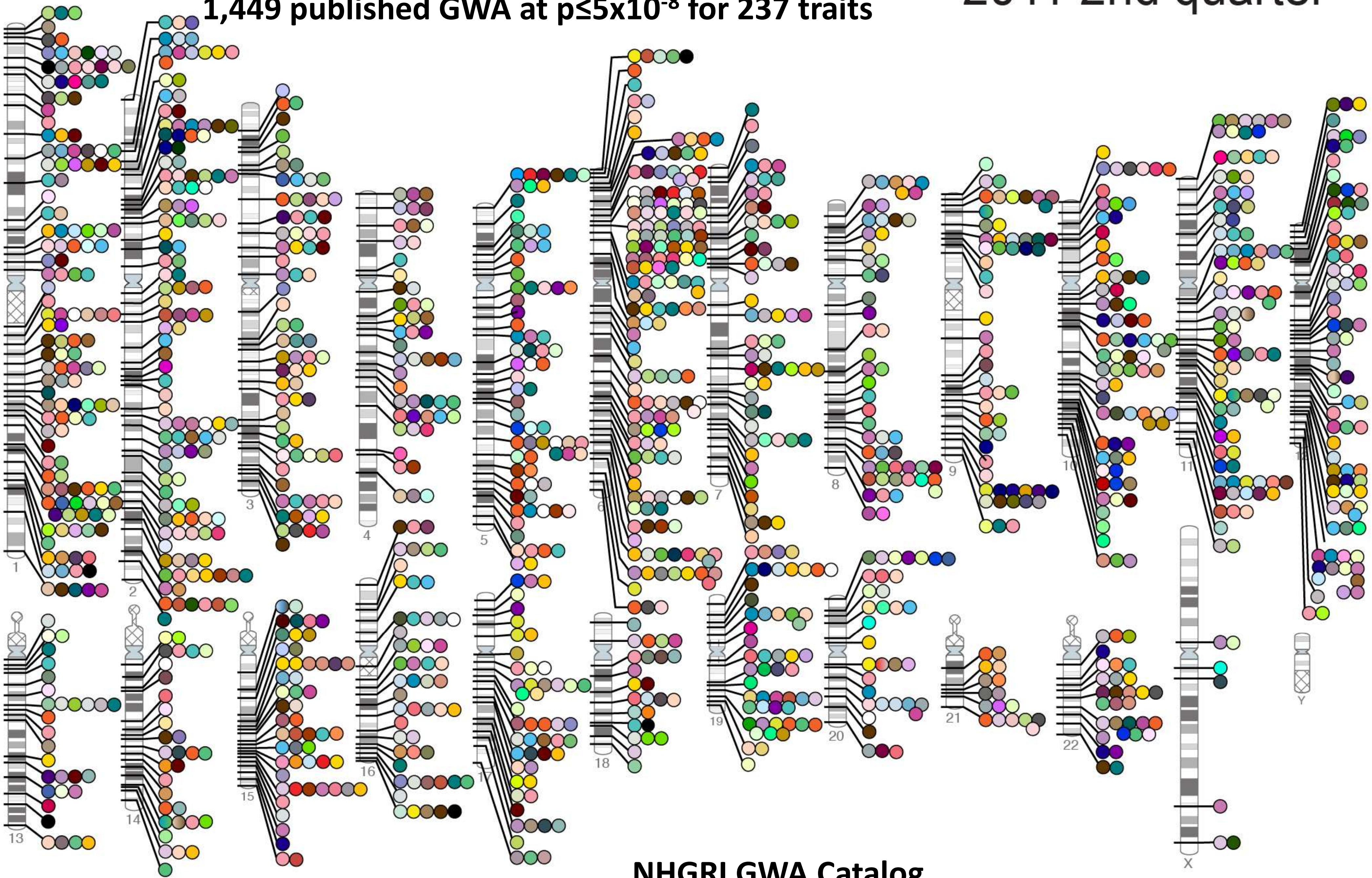
GWAS for Crohn's disease



data from the study of Crohn's disease by the Wellcome Trust Case Control Consortium

Published Genome-Wide Associations through 06/2011, 1,449 published GWA at p≤5x10^{-8} for 237 traits

2011 2nd quarter



NHGRI GWA Catalog
www.genome.gov/GWAStudies

Abdominal aortic aneurysm
Acute lymphoblastic leukemia
Adhesion molecules
Adiponectin levels
Age-related macular degeneration
AIDS progression
Alcohol dependence
Alopecia areata
Alzheimer disease
Amyloid A levels
Amyotrophic lateral sclerosis
Angiotensin-converting enzyme activity
Ankylosing spondylitis
Arterial stiffness
Asparagus anosmia
Asthma
Atherosclerosis in HIV
Atrial fibrillation
Attention deficit hyperactivity disorder
Autism
Basal cell cancer
Behcet's disease
Bipolar disorder
Biliary atresia
Bilirubin
Bitter taste response
Birth weight
Bladder cancer
Bleomycin sensitivity
Blond or brown hair
Blood pressure
Blue or green eyes
BMI, waist circumference
Bone density
Breast cancer
C-reactive protein
Calcium levels
Cardiac structure/function
Cardiovascular risk factors
Carnitine levels
Carotenoid/tocopherol levels
Celiac disease
Celiac disease and rheumatoid arthritis
Cerebral atrophy measures
Chronic lymphocytic leukemia
Chronic myeloid leukemia
Cleft lip/palate
Coffee consumption
Cognitive function
Conduct disorder
Colorectal cancer
Corneal thickness
Coronary disease
Creutzfeldt-Jakob disease
Crohn's disease
Crohn's disease and celiac disease
Cutaneous nevi
Cystic fibrosis severity
Dermatitis
DHEA-s levels
Diabetic retinopathy
Dilated cardiomyopathy
Drug-induced liver injury
Drug-induced liver injury (amoxicillin-clavulanate)
Endometrial cancer
Endometriosis
Eosinophil count
Eosinophilic esophagitis
Erectile dysfunction and prostate cancer treatment
Erythrocyte parameters
Esophageal cancer
Essential tremor
Exfoliation glaucoma
Eye color traits
F cell distribution
Fibrinogen levels
Folate pathway vitamins
Follicular lymphoma
Fuch's corneal dystrophy
Freckles and burning
Gallstones
Gastric cancer
Glioma
Glycemic traits
Hair color
Hair morphology
Handedness in dyslexia
HDL cholesterol
Heart failure
Heart rate
Height
Hemostasis parameters
Hepatic steatosis
Hepatitis
Hepatocellular carcinoma
Hirschsprung's disease
HIV-1 control
Hodgkin's lymphoma
Homocysteine levels
Hypospadias
Idiopathic pulmonary fibrosis
IFN-related cytopeni
IgA levels
IgE levels
Inflammatory bowel disease
Insulin-like growth factors
Intracranial aneurysm
Iris color
Iron status markers
Ischemic stroke
Juvenile idiopathic arthritis
Keloid
Kidney stones
LDL cholesterol
Leprosy
Leptin receptor levels
Liver enzymes
Longevity
LP (a) levels
LpPLA(2) activity and mass
Lung cancer
Magnesium levels
Major mood disorders
Malaria
Male pattern baldness
Mammographic density
Matrix metalloproteinase levels
MCP-1
Melanoma
Menarche & menopause
Meningococcal disease
Metabolic syndrome
Migraine
Moyamoya disease
Multiple sclerosis
Myeloproliferative neoplasms
Myopia (pathological)
N-glycan levels
Narcolepsy
Nasopharyngeal cancer
Natriuretic peptide levels
Neuroblastoma
Nicotine dependence
Obesity
Open angle glaucoma
Open personality
Optic disc parameters
Osteoarthritis
Osteoporosis
Otosclerosis
Other metabolic traits
Ovarian cancer
Pancreatic cancer
Pain
Paget's disease
Panic disorder
Parkinson's disease
Periodontitis
Peripheral arterial disease
Personality dimensions
Phosphatidylcholine levels
Phosphorus levels
Photic sneeze
Phytosterol levels
Platelet count
Polycystic ovary syndrome
Primary biliary cirrhosis
Primary sclerosing cholangitis
PR interval
Progranulin levels
Progressive supranuclear palsy
Prostate cancer
Protein levels
PSA levels
Psoriasis
Psoriatic arthritis
Pulmonary funct. COPD
QRS interval
QT interval
Quantitative traits
Recombination rate
Red vs.non-red hair
Refractive error
Renal cell carcinoma
Renal function
Response to antidepressants
Response to antipsychotic therapy
Response to carbamazepine
Response to clopidogrel therapy
Response to hepatitis C treat
Response to interferon beta therapy
Response to metaformin
Response to statin therapy
Restless legs syndrome
Retinal vascular caliber
Rheumatoid arthritis
Ribavirin-induced anemia
Schizophrenia
Serum metabolites
Skin pigmentation
Smoking behavior
Speech perception
Sphingolipid levels
Statin-induced myopathy
Stroke
Sudden cardiac arrest
Suicide attempts
Systemic lupus erythematosus
Systemic sclerosis
T-tau levels
Tau AB1-42 levels
Telomere length
Testicular germ cell tumor
Thyroid cancer
Thyroid volume
Tooth development
Total cholesterol
Triglycerides
Tuberculosis
Type 1 diabetes
Type 2 diabetes
Ulcerative colitis
Urate
Urinary albumin excretion
Urinary metabolites
Uterine fibroids
Venous thromboembolism
Ventricular conduction
Vertical cup-disc ratio
Vitamin B12 levels
Vitamin D insuffency
Vitiligo
Warfarin dose
Weight
White cell count
White matter hyperintensity
YKL-40 levels

GWAS studies in prostate cancer

GWA studies in cancer

Disease/trait	Sample size Initial	Replication	Region	Gene	Strongest SNP-risk allele	Risk allele frequency in controls	*P*	OR per copy or for heterozygote (95% CI)	Platform manufacturer and SNPs[A]
Prostate cancer (66)	1,453 cases 3,064 con	1,210 cases[B] 2,445 con[B]	8q24.21	Intergenic	rs16901979-A HapC	0.02 0.02	1×10^{-12} 3×10^{-15}	1.79 (1.53–2.11) 2.10 (1.75–2.53)	Illumina: 316,515
Prostate cancer (67)	1,172 cases 1,157 con	3,124 cases 3,142 con	8q24.21	Intergenic	rs6983267-G	0.50	9×10^{-13}	1.26 (1.13–1.41)	Illumina: 538,548
Prostate cancer (108)	1,501 cases 11,290 con	1,992 cases 3,058 con	17q12 17q24.3	*TCF2* Gene-poor region	rs4430796-A rs1859962-G	0.49 0.46	1×10^{-11} 3×10^{-10}	1.22 (1.15–1.30) 1.20 (1.14–1.27)	Illumina: 310,520
Breast cancer (94)	390 cases 364 con	26,646 cases 24,889 con	10q26.13 16q12.1	*FGFR2* *TNCR9/ LOC643714*	rs2981582-G rs3803662-C	0.38 0.25	2×10^{-76} 1×10^{-36}	1.26 (1.23–1.30) 1.20 (1.16–1.24)	Perlegen: 205,586
			5q11.2 8q24.21 11p15.5	*MAP3K1* Intergenic *LSP1*	rs889312-A rs13281615-T rs3817198-T	0.28 0.40 0.30	7×10^{-20} 5×10^{-12} 3×10^{-9}	1.13 (1.10–1.16) 1.08 (1.05–1.11) 1.07 (1.04–1.11)	
Breast cancer	1,145 cases	1,176 cases	10q26.13	*FGFR2*	rs1219648-G	0.40	1×10^{-10}	1.20 (1.07–1.42)	Illumina:

GWAS studies in type I diabetes

Disease/trait	Sample size Initial	Sample size Replication	Region	Gene	Strongest SNP-risk allele	Risk allele frequency in controls	*P*	OR per copy or for heterozygote (95% CI)	Platform manufacturer and SNPs[A]
Type 1 diabetes (105)	1,963 cases 2,938 con	4,000 cases 5,000 con 2,997 trios	12q24.13	*C12orf30*	rs17696736-G	0.42	2×10^{-16}	1.22 (1.15–1.28)	See ref. 35
			12q13.2	*ERBB3*	rs2292239-A	0.34	2×10^{-20}	1.28 (1.21–1.35)	
			16p13.13	*KIAA0350*	rs12708716-A	0.68	3×10^{-18}	1.23 (1.16–1.30)	
			18p11.21	*PTPN2*	rs2542151-C	0.16	1×10^{-14}	1.30 (1.22–1.40)	
			18q22.2	*CD226*	rs763361-A	0.47	1×10^{-8}	1.16 (1.10–1.22)	
Type 1 diabetes (35)	1,963 cases 2,938 con	See ref. 105	12q13.2	*ERBB3*	rs11171739-C	0.42	1×10^{-11}	1.34 (1.17–1.54)	Affymetrix: 469,557
			12q24.13	*SH2B3/LNK*	rs17696736-G	0.42	2×10^{-14}	1.34 (1.16–1.53)	
			16q13.13	*TRAFD1, PTPN11, KIAA0350*	rs12708716-A	0.65	5×10^{-7}	1.19 (0.97–1.45)	
Type 1 diabetes (106)	563 cases 1,146 con 483 trios	2,350 individ in 549 families; 390 trios	16p13.13	*KIAA0350*	rs2903692-G	0.62	7×10^{-11}	1.54 (1.32–1.79)	Illumina: 534,071
Type 2 diabetes (78)	1,380 cases 1,323 con	2,617 cases 2,894 con	8q24.11	*SLC30A8*	rs13266634-C	0.30	6×10^{-8}	1.18 (NR)	Illumina: 392,935
Type 2 diabetes (61)	1,161 cases 1,174 con	1,215 cases 1,258 con	3q27.2	*IGF2BP2*	rs4402960-T	0.30	9×10^{-16}	1.14 (1.11–1.18)	Illumina: 315,635
			6p22.3	*CDKAL1*	rs7754840-C	0.36	4×10^{-11}	1.12 (1.08–1.16)	
			9p21.3	*CDKN2A/B*	rs10811661-T	0.85	8×10^{-15}	1.20 (1.14–1.25)	
			11p12	Intergenic	rs9300039-C	0.89	4×10^{-7}	1.25 (1.15–1.37)	
Type 2 diabetes (90)	1,464 cases 1,467 con	5,065 cases 5,785 con	9p21.3	*CDKN2A/B*	rs10811661-T	0.83	8×10^{-15}	1.20 (1.14–1.25)	Affymetrix: 386,731
			3q27.2	*IGF2BP2*	rs4402960-T	0.29	9×10^{-16}	1.14 (1.11–1.18)	
			6p22.3	*CDKAL1*	rs7754840-C	0.31	4×10^{-11}	1.12 (1.08–1.16)	
Type 2 diabetes (107)	1,399 cases[B] 5,275 con[B]	2,437 cases[B] 7,287 con[B]	6p22.3	*CDKAL1*	rs7756992-G	0.26	8×10^{-9}	1.20 (1.13–1.27)	Illumina: 313,179 SNPs;

GWAS studies in eye diseases

GWA studies in eye diseases

Disease/trait	Sample size Initial	Replication	Region	Gene	Strongest SNP-risk allele	Risk allele frequency in controls	*P*	OR per copy or for heterozygote (95% CI)	Platform manufacturer and SNPs[A]
Age-related macular degeneration (63)	96 cases 50 con	NR	1q31	*CFH*	rs380390-C	0.70 (HapMap CEU)[B]	4×10^{-8}	4.6 (2.0–11)	Affymetrix: 103,611
Wet age-related macular degeneration (95)	96 cases 130 con	NR	10q26.13	*HTRA1*	rs11200638-A	NR	8×10^{-12}	1.60 (0.71–3.61)	Affymetrix: 97,824
Exfoliation glaucoma (65)	75 cases 14,474 con	254 cases 198 con	15q24.1	*LOXL1*	rs3825942-G	0.85	3×10^{-21}	20.10 (10.80–37.41)	Illumina: 304,250

HapMap historical timeline

Project Launched – October 2001

Limitations of HapMap-based GWA studies

Not optimal for assessing disease associations with rare variants
Not optimal for assessing disease associations with structural variants, such as copy number polymorphisms, insertions, deletions, and inversions
Requires very large number of samples
Patterns of genomic variation in populations other than the original HapMap samples may not be optimally described

- Published 2007

HapMap 3 – data generation 2008, published 2010

- Used Affy 6.0 & Illumina 1M to expand to >1200 samples

@Daly, M

Key lessons from GWAS

Functional analysis

Based on a complete catalogue of SNP-trait associations in human

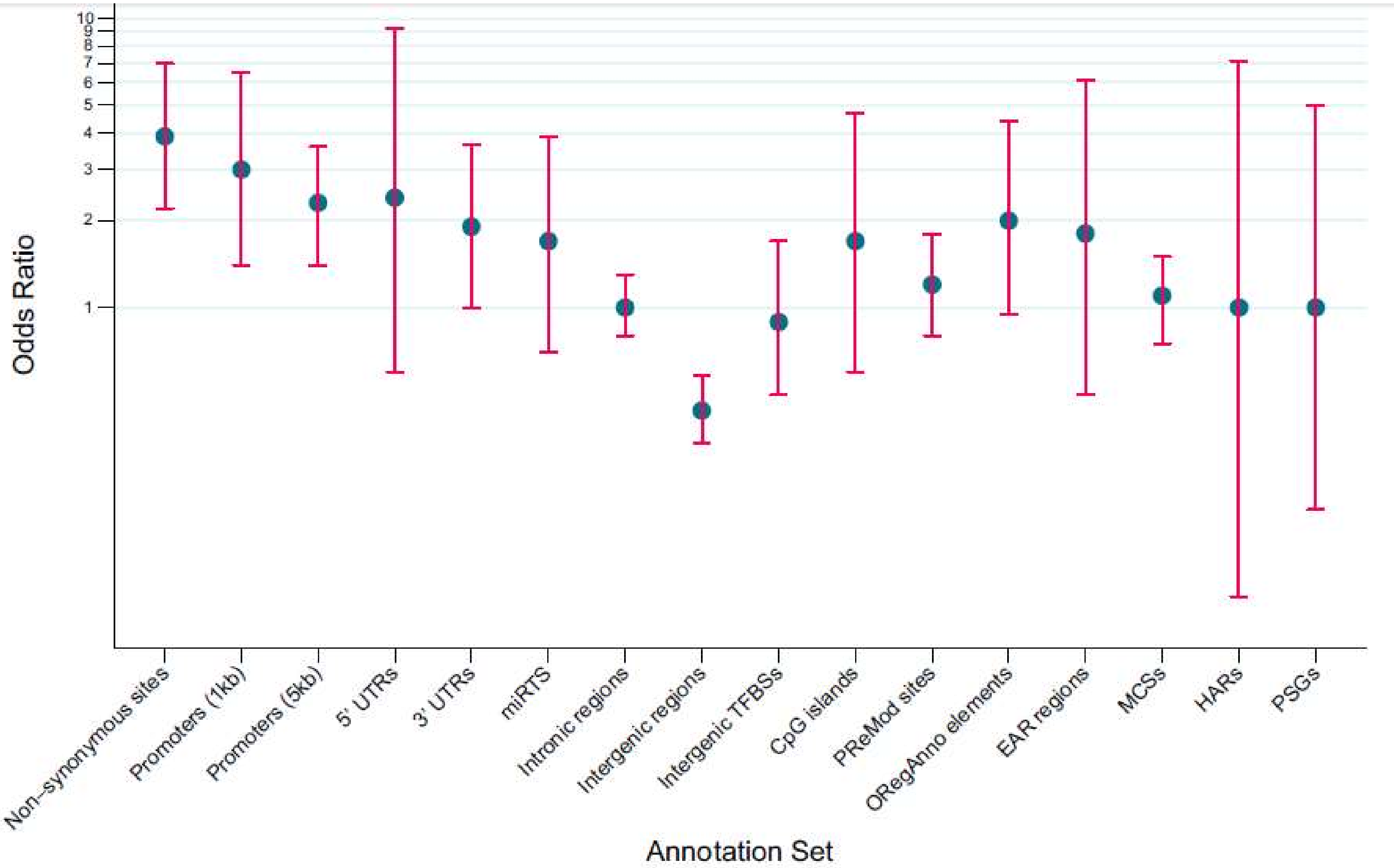
Annotation set	Description	Source	Frequency (n, %) of TAS blocks*	Frequency of Random LD blocks*
Non-synonymous	Genomic positions wherein a nucleotide substitution would cause an amino acid replacement	dbSNP version 129	57, 12.2%	15.8, 3.5%
Promoters (1kb)	1kb regions upstream of annotated transcription start sites	Ensembl v49 release at www.ensembl.org	26, 5.5%	8.8, 1.9%
Promoters (5kb)	5kb regions upstream of annotated transcription start sites	Ensembl v49 release at www.ensembl.org	65, 13.9%	30.3, 6.7%
5′ UTR	5′ untranslated regions	dbSNP version 129	7, 1.5%	2.8, 0.6%
3′ UTR	3′ untranslated regions	dbSNP version 129	28, 6.0%	14.6, 3.2%
Intronic	Non-coding regions within a gene	dbSNP version 129	189, 40.3%	183.2, 40.3%
Splice sites	Intronic regions that allow for splicing (removing introns and joining exons)	dbSNP version 129	0, 0%	0.1, 0.02%
Intergenic	Non-coding regions outside of genes	dbSNP version 129	186, 40.0%	276.0, 60.7%

- The vast majority of associated variants fall outside coding regions
- over-representation of non-synonymous associated variants
- over-representation of associated variants in promoters

Hindorff PNAS 2009

Functional analysis

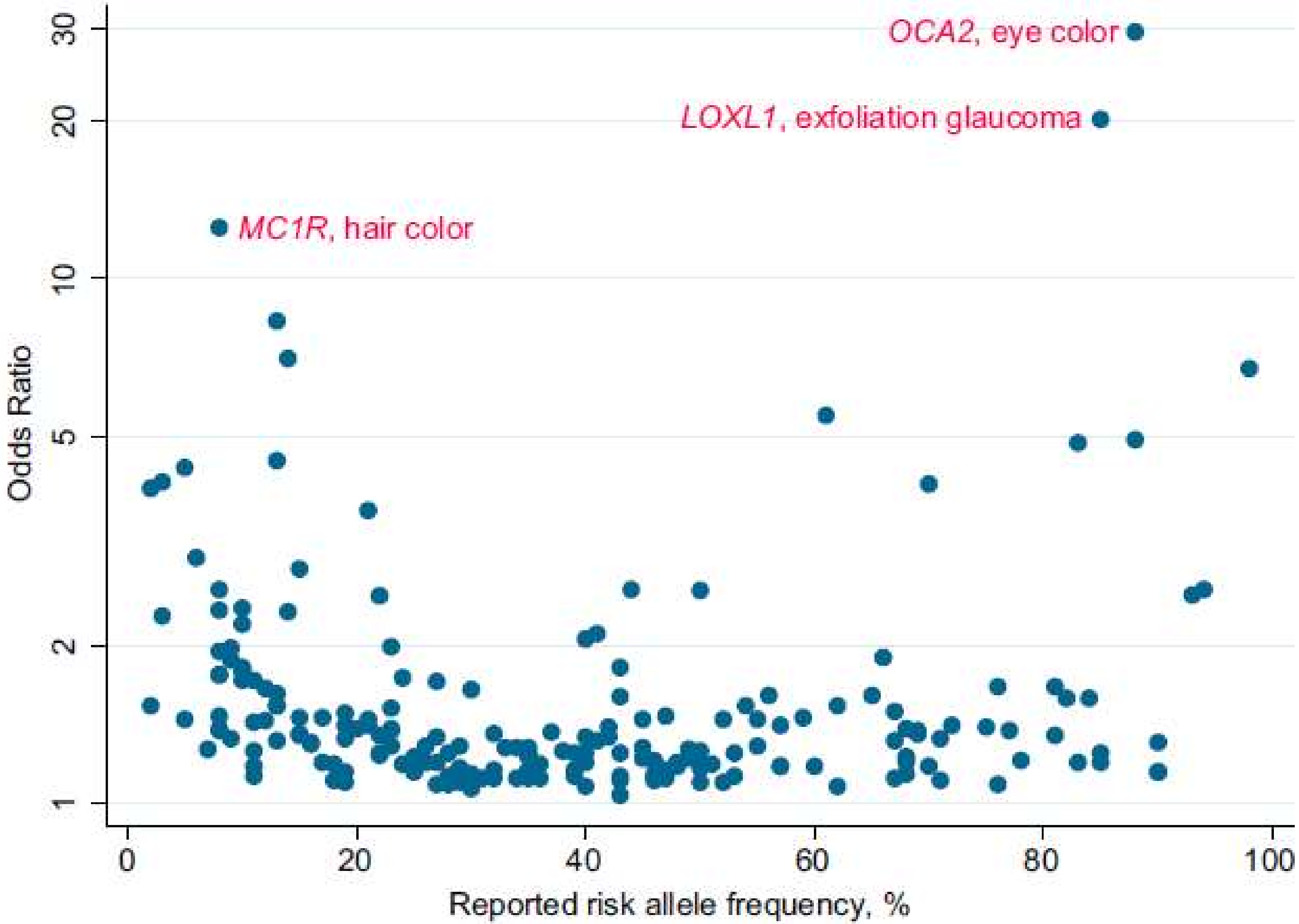
Based on a complete catalogue of SNP-trait associations in human



- Disease associated variants in coding regions have the highest effect size (OR)
- Strong signal also in promoter regions

Hindorff PNAS 2009

Functional analysis



Relatively modest effect size observed for genetic variants associated with common diseases.

Hindorff PNAS 2009

Genetic odds ratio

- Groups = genotypes,
- an event = case (disease)

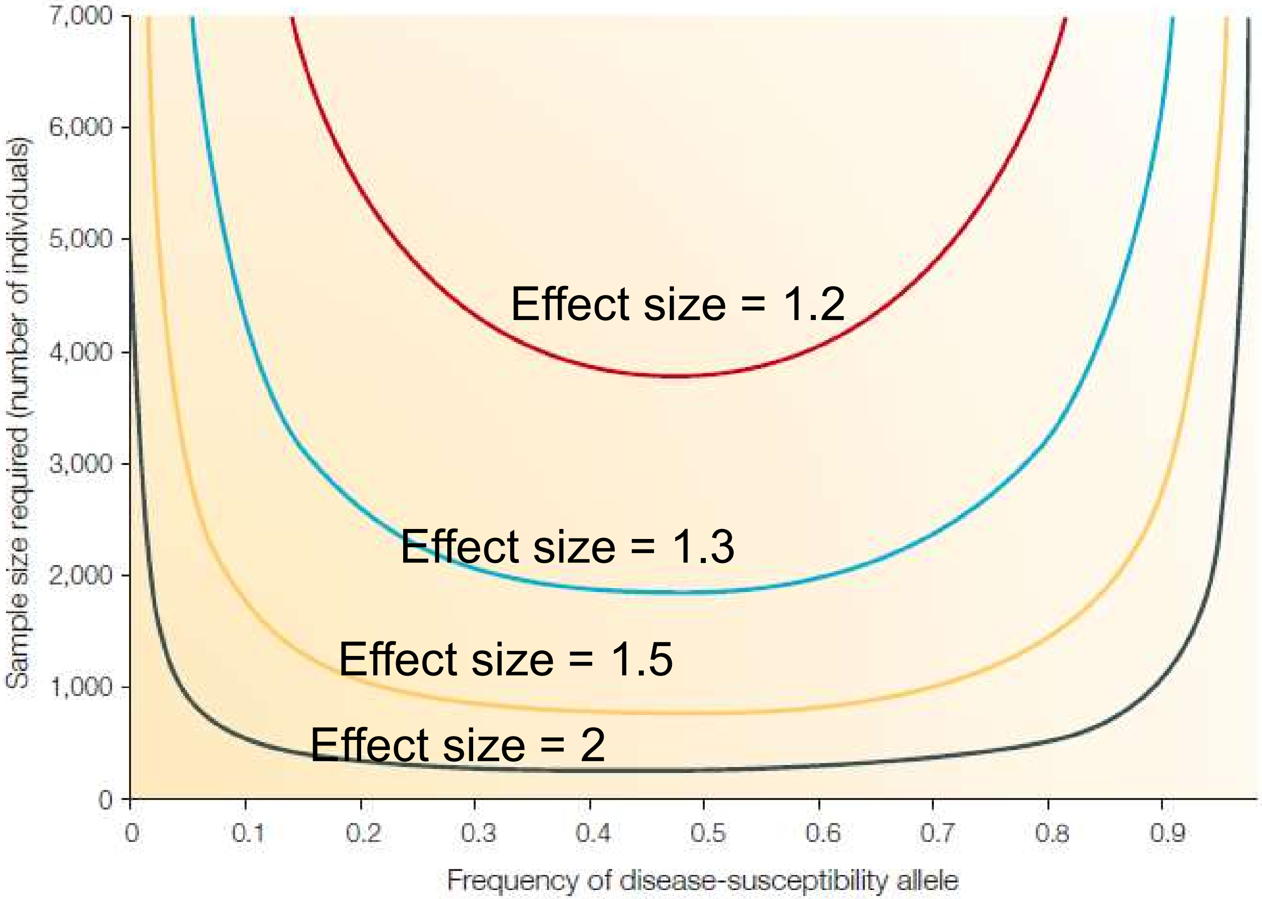
	AA	AB
Case	r_0	r_1
Control	s_0	s_1

$$OR = \frac{r1/s1}{r0/s0}$$

Effects of allele frequency on sample-size requirements

$$\mathbb{E}[\chi^2] \propto N\gamma^2 p(1-p)r^2,$$

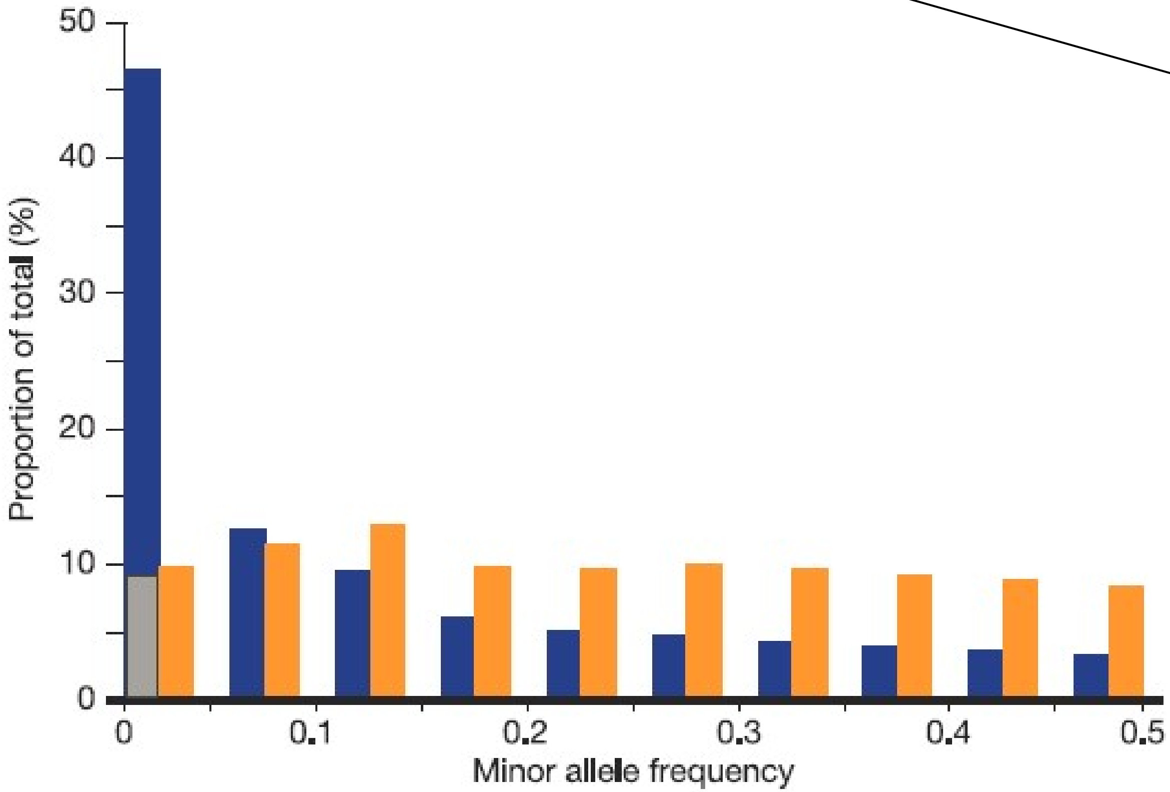
where χ^2 is the chi-squared test statistic, N the number of cases and controls γ the effect size, p the allele frequency of the risk variant and r^2 is the correlation between the marker and causal SNP.



Relatively modest effect size: A consequent need for large samples

Wang et al. 2005

Minor allele frequency (MAF) distribution in the ENCODE data

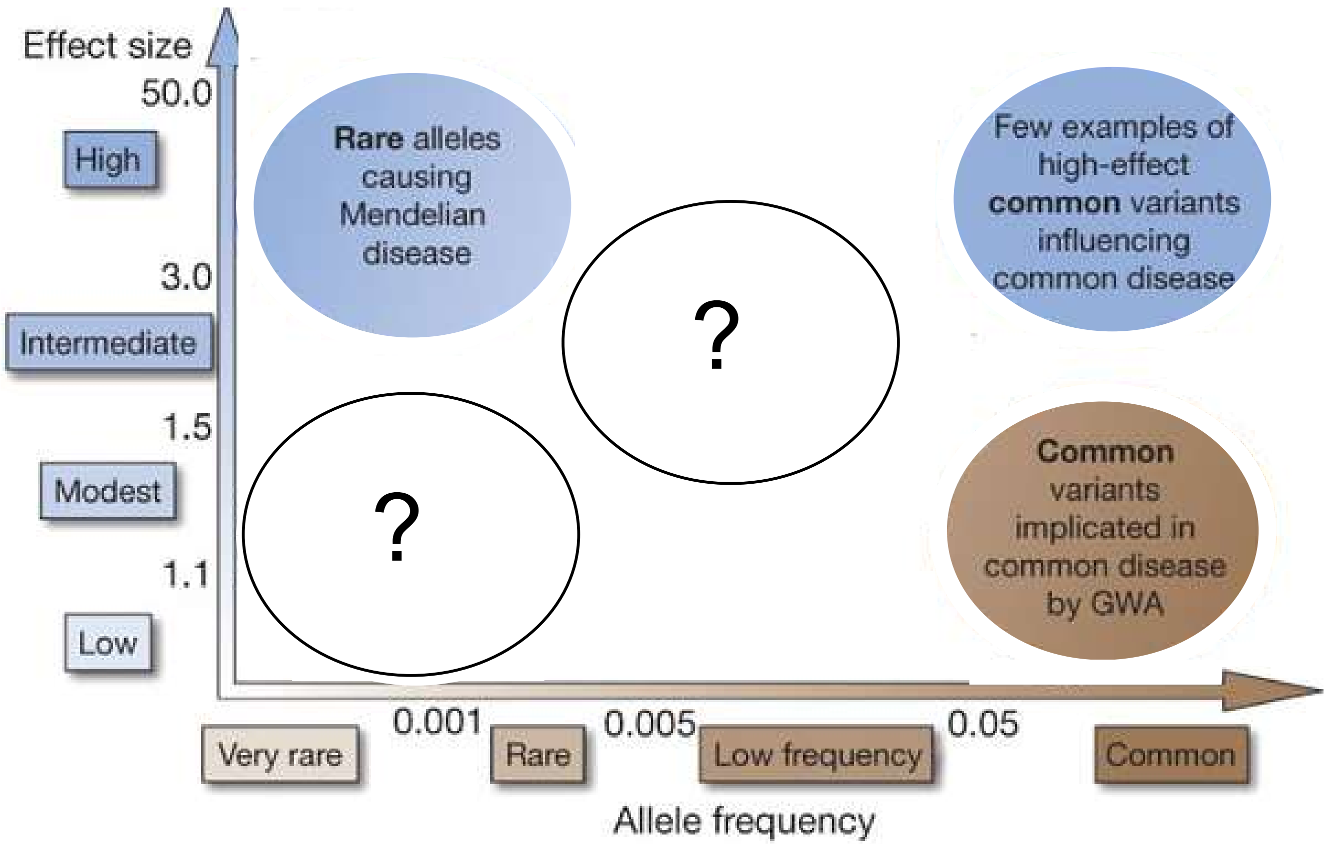


10 regions, each of 500kb, sequenced in 269 DNA samples.

Polymorphic SNPs in population

heterozygosity in individuals

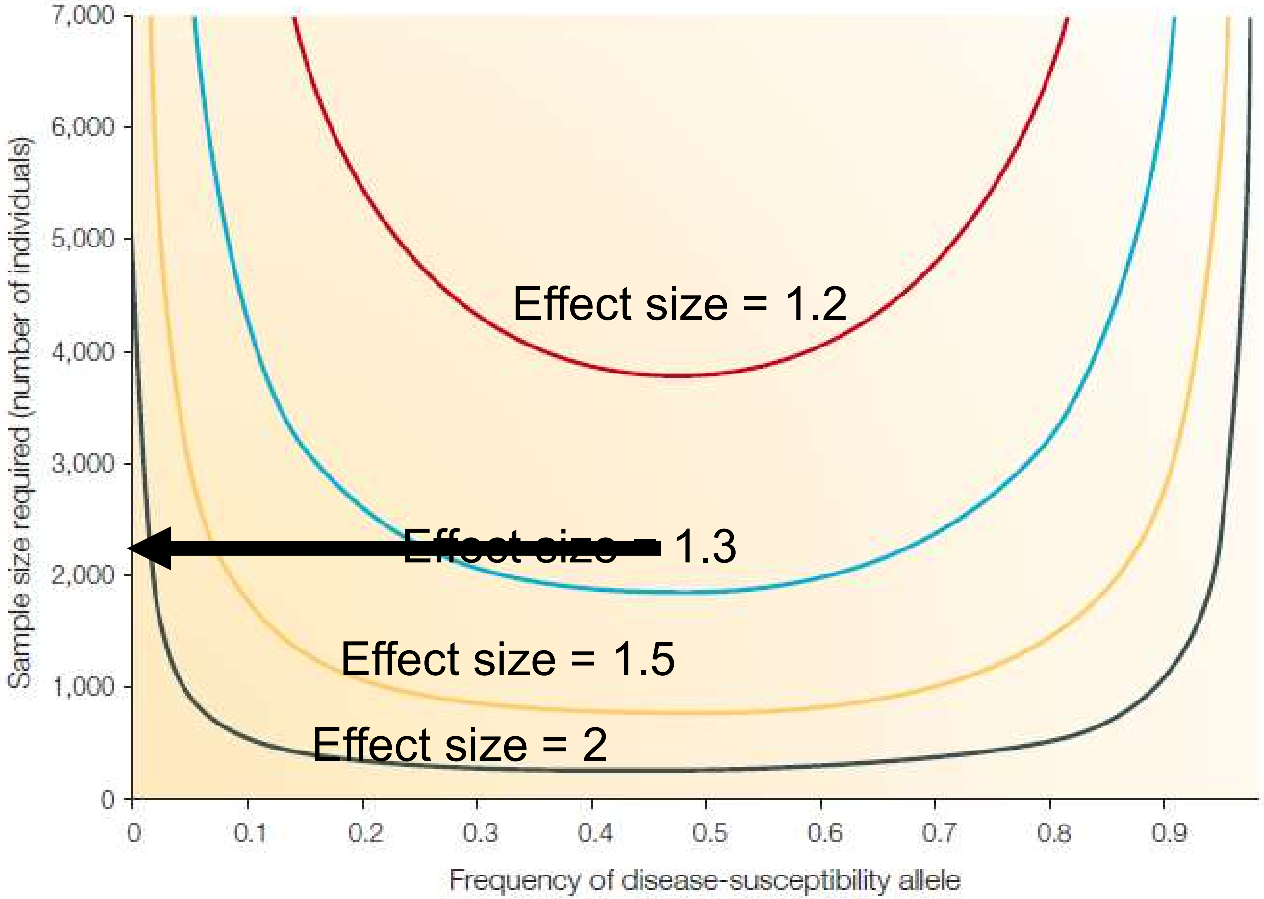
Feasibility of identifying variants



Effects of allele frequency on sample-size requirements

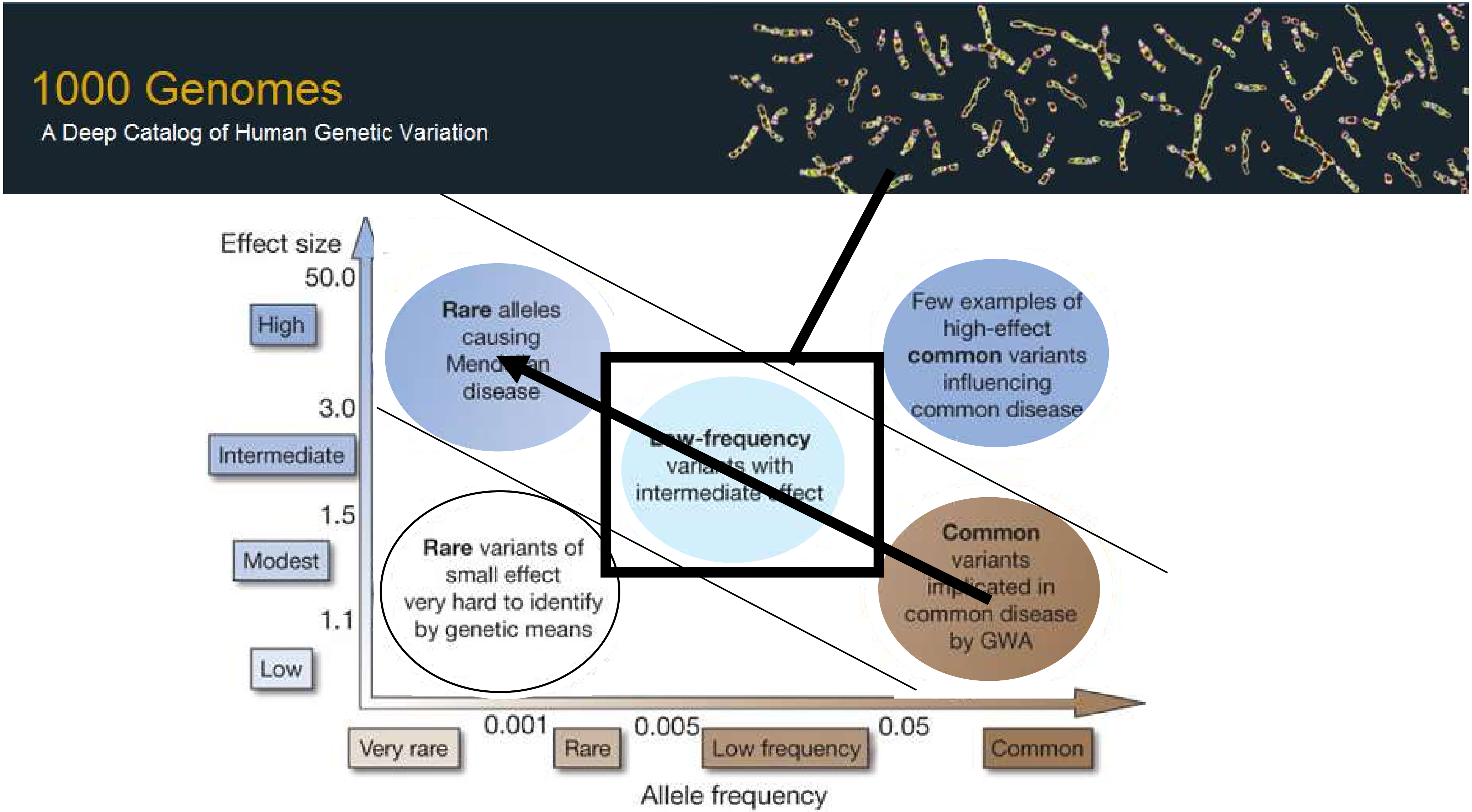
$$\mathbb{E}[\chi^2] \propto N\gamma^2 p(1-p)r^2,$$

where χ^2 is the chi-squared test statistic, N the number of cases and controls γ the effect size, p the allele frequency of the risk variant and r^2 is the correlation between the marker and causal SNP.



Wang et al. 2005

Feasibility of identifying associations

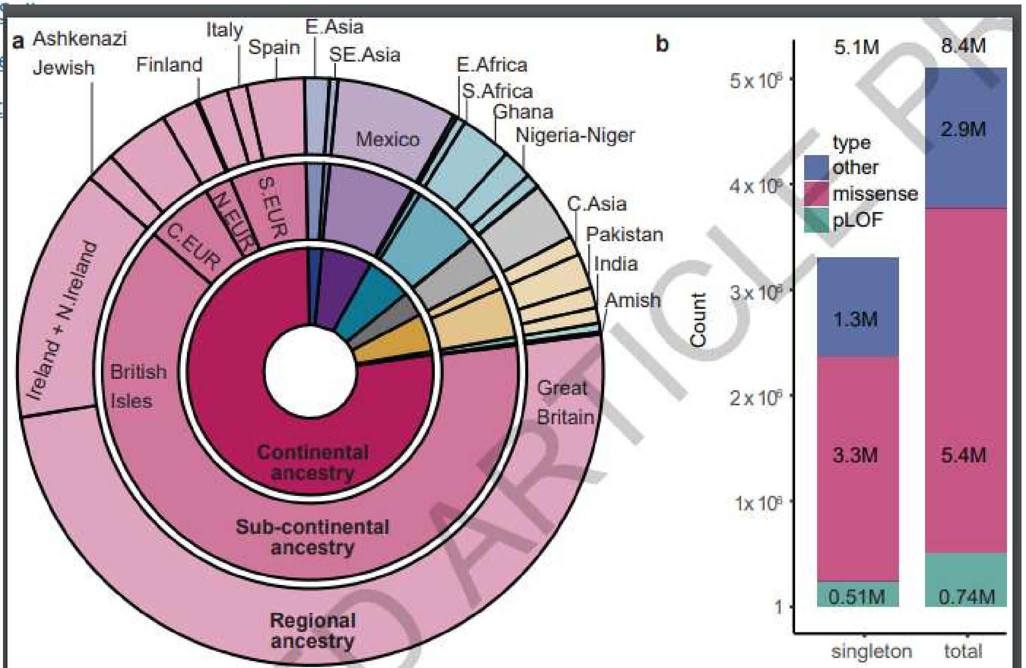


Article | Published: 20 May 2024

A deep catalogue of protein-coding variation in 983,578 individuals

Kathie Y. Sun, Xiaodong Bai, Siying Chen, Suying Bao, Chuanyi Zhang, Manav Kapoor, Joshua Backman, Tyler Joseph, Evan Maxwell, George Mitra, Alexander Gorovits, Adam Mansfield, Boris Boutkov, S… Gokhale, Lukas Habegger, Anthony Marcketta, Adam E. Locke, Liron Ganel, Alicia Hawes, Michae… Kessler, Deepika Sharma, Jeffrey Staples, Jonas Bovijn, Sahar Gelfman, Regeneron Genetics Cent… ME Cohort Partners, … Suganthi Balasubramanian

+ Show authors



AFR	49,408	6.0%
EAS	15,006	1.8%
EUR	630,616	76.7%
IAM	54,601	6.6%
MEA	3,903	0.5%
SAS	40,668	5.0%
UNK	27,776	3.4%

c

Functional category	Total variants	Percent singletons	Per-sample counts med
frameshift	560,431	58%	71
splice acceptor	96,080	54%	10
pLOFs	1,115,116	53%	137
splice donor	119,147	51%	10
stop gained	339,463	47%	46
in-frame indel	184,652	46%	148
missense	10,444,562	40%	8,652
synonymous	4,645,092	36%	10,185

(The missing) heritability

Using unbiased estimators to evaluate heritability

Table 19.1 Summary of a one-way ANOVA involving N independent pairs of twins.

Source of Variance	df	Sums of Squares	Observed Mean Squares	$E(\text{MS})$
Among pairs	$N-1$	$\text{SS}_b = 2\sum_{i=1}^{N}(\bar{z}_i - \bar{z})^2$	$\text{SS}_b/(N-1)$	$\sigma_e^2 + 2\sigma_b^2$
Within pairs	N	$\text{SS}_e = \sum_{i=1}^{N}\sum_{j=1}^{2}(z_{ij} - \bar{z}_i)^2$	SS_e/N	σ_e^2
Total	$T-1$	$\text{SS}_T = \sum_{i=1}^{N}\sum_{j=1}^{2}(z_{ij} - \bar{z})^2$	$\text{SS}_T/(T-1)$	σ_z^2

An upper bound on (broad sense) heritability based on monozygotic twins

$$H^2 = \frac{s^2_{among-pairs}}{s^2_{among\_pairs} + s^2_{within-pairs}}$$

Var explained by genetic background

Source of variation		df	SS	MS
$\bar{Y} - \bar{\bar{Y}}$	Among groups	$a - 1$	SS_{among}	$\dfrac{SS_{among}}{a - 1}$
$Y - \bar{Y}$	Within groups	$\sum^{a} n_i - a$	SS_{within}	$\dfrac{SS_{within}}{\sum^{a} n_i - a}$
$Y - \bar{\bar{Y}}$	Total	$\sum^{a} n_i - 1$	SS_{total}	

The **unbiased estimator of** σ_{within}^2 σ_{Among}^2

$$s^2 = MS_{within} \qquad s_A^2 = \frac{MS_{among} - MS_{within}}{n_0}$$

$$n_0 = \frac{1}{a-1}\left(\sum_{i=1}^{a} n_i - \frac{\sum_{i=1}^{a} n_i^2}{\sum_{i=1}^{a} n_i}\right)$$

The missing heritability

Table 1 | GWAS for common diseases and traits

Phenotype	Number of GWAS loci	Proportion of heritability explained (%)*
Type 1 diabetes	41	~60
Fetal haemoglobin levels	3	~50
Macular degeneration	3	~50
Type 2 diabetes	39	20–25
Crohn's disease	71	20–25
LDL and HDL levels	95	20–25
Height	180	~12

HDL, high-density lipoprotein; LDL, low-density lipoprotein.
* Fraction of heritability explained is calculated by dividing the phenotypic variance explained by variants at loci identified by GWAS by the total heritability as inferred from epidemiological parameters.

The case of the missing heritability

When scientists opened up the human genome, they expected to find the genetic components of common traits and diseases. But they were nowhere to be seen. **Brendan Maher** shines a light on six places where the missing loot could be stashed away.

If you want to predict how tall your children might one day be, a good bet would be to look in the mirror, and at your mate. Studies going back almost a century have estimated that height is 80–90% heritable. So if 29 centimetres separate the tallest 5% of a population from the shortest, then genetics would account for as many as 27 of them[1].

This year, three groups of researchers[2-4] scoured the genomes of huge populations (the largest study[4] looked at more than 30,000 people) for genetic variants associated with the height differences. More than 40 turned up.

But there was a problem: the variants had tiny effects. Altogether, they accounted for little more than 5% of height's heritability — just 6 centimetres by the calculations above. Even though these genome-wide association studies (GWAS) turned up dozens of variants, they did "very little of the prediction that you would do just by asking people how tall their parents are", says Joel Hirschhorn at the Broad Institute in Cambridge, Massachusetts, who led one of the studies[3].

Height isn't the only trait in which genes have gone missing, nor is it the most important. Studies looking at similarities between identical and fraternal twins estimate heritability at more than 90% for autism[5] and more than 80% for schizophrenia[6]. And genetics makes a major contribution to disorders such as obesity, diabetes and heart disease. GWAS, one of the most celebrated techniques of the past five years, promised to deliver many of the genes involved (see "Where's the reward?", page 20). And to some extent they have, identifying more than 400 genetic variants that contribute to a variety of traits and common diseases. But even when dozens of genes have been linked to a trait, both the individual and cumulative effects are disappointingly small and nowhere near enough to explain earlier estimates of heritability. "It is the big topic in the genetics of common disease right now," says Francis Collins, former head of the National Human Genome Research Institute (NHGRI) in Bethesda, Maryland. The unexpected results left researchers at a point "where we all had to scratch our heads and say, 'Huh?'", he says.

Although flummoxed by this missing heritability, geneticists remain optimistic that they can find more of it. "These are very early days, and there are things that are doable in the next year or two that may well explain another sizeable chunk of heritability," says Hirschhorn. So where might it be hiding?

Right under everyone's noses

The inability to find some genes could be explained by the limitations of GWAS. These studies have identified numerous one-letter variations in DNA called single nucleotide polymorphisms (SNPs) that co-occur with a disease or other trait in thousands of people. But a given SNP represents a much bigger block of genetic material. So, for example, if two people share one of these variants at a key location, both may be scored as having the same version of any height-related gene in that area, even though one person actually has a relatively rare mutation that has a huge effect on height. The association study might identify a variant responsible for the height difference, says Teri Manolio, director of the Office of Population Genomics at the NHGRI, but averaging across hundreds of people could give the appearance that its effects are pretty wimpy. "It's going to be diluted," she says.

Finding this type of missing heritability is conceptually easy, because it involves closer scrutiny of the genes already in hand. "Just exploring, in a very dense way, genetic variation at the loci that have been discovered is probably going to [explain] another increment of missing heritability," Hirschhorn says. Researchers will need to sequence candidate genes and their surrounding regions in thousands of people if they are to unearth more associations with the disease.

Helen Hobbs and Jonathan Cohen of the University of Texas Southwestern Medical Center in Dallas did this in an attempt to capture all the variation in *ANGPTL4*, a gene their studies had linked to cholesterol and triglyceride concentrations. They sequenced the gene in around 3,500 individuals from the Dallas Heart Study and found that some previously unknown variants had dramatic effects on the concentration of these lipids in the blood[7]. Mark McCarthy of Britain's Oxford Centre for Diabetes, Endocrinology and Metabolism says that such studies could reveal much of the missing heritability, but not a lot of people have had the enthusiasm to do them. This could change as the cost of sequencing falls.

Out of sight

Other variants, for which GWAS haven't even begun to provide clues, will prove even harder to find. In the past, conventional genetic studies for inherited diseases such as cystic fibrosis identified rare, mutated genes that have a high penetrance, meaning that the gene has an effect in almost everyone who carries it. But it quickly became apparent that high-penetrance variants would not underlie most common diseases because evolution largely keeps them in check.

What powered the push into genome-wide association was a hypothesis that common diseases would be caused by common, low-penetrance variants when enough of them showed up in the same unlucky person. Now that hypothesis is being questioned. "A lot of people are recognizing that screening for common variation has delivered less than we had hoped," says David Goldstein, professor of genetics at Duke University in Durham, North Carolina.

But between those variants that stick out like a sore thumb, and those common enough to be dredged up by the wide net of GWAS, there is a potential middle ground of variants that are moderately penetrant but are rare enough that they are missed by the net. There's also the possibility that there are many more-frequent variants that have such a low penetrance that GWAS can't statistically link them to a disease.

These very-low-penetrance variants pose some problems, says Leonid Kruglyak professor of ecology and evolutionary biology at Princeton University in New Jersey. "You're talking about thousands of variants that you would have to invoke to get near 80% or 90% heritability." Taken to the extreme, practically every gene in the genome could have a variant that affects height, for example. "You don't like to think about models like that," Kruglyak says.

If rare, moderately penetrant or common, weakly penetrant variants are the culprits, then bumping up the number of people in existing association studies could help find previously missed genetic associations. Peter Visscher of the Queensland Institute of Medical Research in Brisbane, Australia, says that a meta-analysis of height studies covering roughly 100,000 people is in the works. Lowering the stringency with which an association is made could drag up more, but confidence in the hits would drop.

At some point it might make sense to stop using SNPs, and start sequencing whole genomes. Collins suggests that the NHGRI's 1,000 genomes project, which aims to sequence the genomes of at least 1,000 people from all over the world, could go a long way towards finding hidden heritability, and many more genomes may become possible as the price of sequencing falls.

Not everyone supports an all-out sequencing onslaught. Goldstein warns against

Where's the reward?

There is more riding on the case of missing heritability than academic satisfaction. By finding variants related to common disease, genome-wide association studies promised to deliver meaningful medical information and justify the US$3 billion spent on the human genome and the multimillion-dollar effort to map human variation. "The reason for spending so much money was that the bulk of the heritability would be discovered," says Joseph Nadeau, a geneticist at Case Western Reserve University in Cleveland, Ohio.

The ability to predict someone's height from their genes would be a pretty trivial carnival trick, but it represents a mastery over the language of life that could potentially spill into most areas of medicine. Aside from some surprises, though, such as mutations in immune-system genes being tied to an eye disorder called age-related macular degeneration, many of the variants found have only modest effects on human characteristics. For now, genetics rarely provides a clearer predictive answer than a good family history. And the path to therapy is not straightforward, says David Goldstein of Duke University in Durham, North Carolina. "This talk about personalized risk profiles, using genetics, for most common diseases, and this talk about a whole flood of new drug targets. I think that that's now pretty clearly wishful thinking."

Francis Collins, former head of the National Human Genome Research Institute in Bethesda, Maryland, agrees that the picture for disease prediction remains bleak, but is still optimistic about therapeutic intervention. Most genetic variants found by genome-wide association "contribute a relatively modest risk, but that in no way says the genes aren't important," he says. "The opportunity for therapy here is breathtaking."

Peter Visscher, a geneticist at the Queensland Institute of Medical Research in Brisbane, Australia, agrees. "It would be easy to knock [genome-wide association studies] and say everything was promised and nothing was delivered. But in terms of identifying genes and pathways for disease, it's been very successful. I would feel it's moved the field forwards tremendously."

Ultimately, the clinical value of the variants that genome-wide association studies have turned up may differ from disease to disease. Still, some say that the field is too fixated on clinical application, be it through prediction, personalization or identifying drug targets. Robert Nussbaum of the University of California, San Francisco, puts it bluntly: "Human genetics research always assumes too quickly that it has to be translational. They're doing basic research." B.M.

continuing to "turn the crank" without devising a more rational approach, such as sequencing the genomes of people who exhibit extreme manifestations of diseases. "I'm not really sold on doing the sequencing version of what we did with [GWAS]," he says. "It's a big enough, costly enough job, that I think we want to think a little bit harder about exactly who gets re-sequenced."

In the architecture

Some researchers are now homing in on copy-number variations (CNVs), stretches of DNA tens or hundreds of base pairs long that are deleted or duplicated between individuals. Variations in these features could begin to explain missing heritability in disorders such as schizophrenia and autism, for which GWAS have turned up almost nothing. Two recent studies looked at hundreds of CNVs in normal people and in those with schizophrenia, and found strong associations between the disease and several CNVs[8,9]. They commonly arise *de novo* — in an individual without any family history of the mutation.

These structural variants might account for a lot of the genetic variability from person to person and could account for some of those rare 'out-of-sight' mutations with moderate penetrance that GWAS can't pick up. Many CNVs go undetected because they don't alter SNP sequences. Duplicated regions can also be difficult to sequence.

A standard technology for uncovering CNVs is array comparative genomic hybridization, in which scientists examine how genetic material from different individuals hybridizes to a microarray. If certain spots on an array pick up more or less DNA, it could indicate that there's a CNV. This and several other techniques are being tested by a consortium called the Copy Number Variation Project, run out of the Wellcome Trust Sanger Institute in Cambridge, UK. The consortium is dedicated to characterizing as many CNVs as possible so that associations can be made between them and diseases. McCarthy says that the role hidden CNVs have in heritability "should play out in the next six months to a year". But Goldstein argues that current technologies will miss many of the smaller CNVs, from 50 base pairs down to repeats of just two bases. "All we'll have verification of is the big whopping CNVs that are identifiable, and they clearly do not account for much of the missing heritability."

In underground networks

Most genes work together with close partners, and it is possible that the effects of one on heritability cannot be found without knowing the effects of the others. This is an example of epistasis, in which one gene masks the effect of another, or where several genes work together. Two genes may each add a centimetre to height on their own, for example, but together they could add five. GWAS don't cope with epistasis very well, and efforts to find these interactions usually require good up-front guesses about the interacting partners.

Joseph Nadeau, a geneticist at Case Western Reserve University in Cleveland, Ohio, says that 'modifier' genes act even in some straightforward single-gene diseases. "That's a simple kind of epistasis," he says. Cystic fibrosis, for example, is usually caused by mutations in one gene, *CFTR*, yet can vary greatly in symptoms and severity. The suspicion has been that modifier genes are one cause of this variability.

But despite the years of study, researchers still struggle to pin down these genes. "People haven't modelled truly the effect of epistasis," says population geneticist Sarah Tishkoff at the University of Pennsylvania in Philadelphia.

It's no surprise that genetics is more complicated than one gene, one phenotype, or even several genes, one phenotype, but it's humbling to realize how much more complex things are starting to look. In a now classic study[10], Kruglyak and his colleagues found that expression of most yeast genes is controlled by several variants, often more than five. To fill in all the heritability blanks, researchers may need better and more varied models of the entire network of genes and regulatory sequences, and of how they act together to produce a phenotype. At some point this process starts to look more like systems biology, and researchers are already applying systems methods to humans and other organisms (see page 26). "What we're learning from these studies is that we need to think about the more complex of the complex models rather than the more simple of the complex models," Kruglyak says.

The great beyond

What if heritability estimates were wrong in the first place? Heritability of height was initially measured by taking the mean height of parents and comparing that value to the adult height of their offspring. As the average heights of parents increase, researchers found, so too does the average height of their children, hence the calculated 80–90% heritability.

Environment, especially factors such as nutrients or toxins present during important growth phases, can affect the mean height of a population considerably — but researchers have controlled for environment in estimates of heritability by, for example, comparing genetically identical twins raised together with those raised apart. Most researchers are confident that the heritability estimates are sound. "I don't think anyone's going to say that the heritability of height is 10% and let environment get you closer to the answer," Kruglyak says. "I don't think you can explain it away."

But there are lingering doubts about how precisely environment has been accounted for in heritability studies. Adverse experiences *in utero* could lead to lifelong health disparities, according to David Barker from the University of Southampton, UK, and yet a shared womb is an aspect of the environment that would not be factored into such studies. "Heritability estimates are basically what clusters in families, and environment clusters in families," says Manolio.

Epigenetics, changes in gene expression that are inherited but not caused by changes in genetic sequence, confuses things further. Feeding a mouse a certain diet, for example, can alter the coat colour not only in its children, but also in its children's children[11]. Here, the expression of a coat-colour gene is controlled by a type of DNA modification called methylation, but it's not completely clear how that methylation pattern is 'remembered' by the next generation. The idea that grandma's environment could affect future generations is controversial — and such effects would have been included in the heritability normally attributed to genes.

"This complicates everything," says Nadeau. "How do we sort out what great-grandfather and great-grandmother were exposed to when they were young and having children?" Model organisms might help. Nadeau has investigated testicular germ-cell tumours in mice that are analogous to a highly heritable cancer in humans. His group found that the effects of one weak, cancer-promoting gene, *Dnd1*^{Ter}, are greatly enhanced by several other gene variants, and the boosted effects are passed on even if the genes that cause them are not[12].

"It's presumably transmitting its presence in some epigenetic way," says Nadeau. The mechanisms by which epigenetic inheritance might work are still disputed, though; marks such as methylation that direct gene expression during someone's life seem to be wiped clean in a new embryo. One possible explanation for Nadeau's observation, he says, is that RNA is being inherited alongside DNA through sperm or eggs.

Collins is not convinced that epigenetics will play a big part in missing heritability in humans. "It just doesn't look likely outside of one or two examples to suggest that this is the case." Nadeau disagrees. "It's hard to imagine that every other organism works one way and humans are the exception," he says.

Lost in diagnosis

There is a nagging worry as researchers hunt for heritability: that common diseases might not, in fact, be common. Medicine tries hard to lump together a complex collection of symptoms and call it a disease. But if thousands of rare genetic variants contribute to a single disease, and the genetic underpinnings can vary radically for different people, how common is it? Are these, in fact, different diseases?

GWAS could actually be proving so difficult because researchers are seeking shared susceptibility genes in a group of people who may share few, if any. And yet without a more refined understanding of genetics, it could be impossible to categorize them any better. "It may be rare variants, common disease. And that's kind of scary to people because it's much, much harder to find those," says Tishkoff.

There could be scarier and more intractable reasons for unaccounted-for heritability that are not even being discussed. "It's a possibility that there's something we just don't fundamentally understand," Kruglyak says. "That it's so different from what we're thinking about that we're not thinking about it yet."

Still the mystery continues to draw its sleuths, for Kruglyak as for many other basic-research scientists. "You have this clear, tangible phenomenon in which children resemble their parents," he says. "Despite what students get told in elementary-school science, we just don't know how that works." ■

Brendan Maher is a Features editor for *Nature*.

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See Editorial, page 1, and News Feature, page 26.

Possible explanations

- GWAS miss common variants of smaller effect size due to limited sample size
- GWAS does not measure non-common variants
- GWAS miss common variants that are in low LD with measured variants
- Heritability measure may be inflated

Additional explanations

1. Rare variants
2. Structural diversity
3. Gene-gene epistasis
4. Gene-environment interaction
5. Transgenerational effects (epigenetics)
6. Transgenerational effects (microRNAs)
7. Parental origin of sequence variants